Two Approximations Under One Name: Separating Sobolev Line Transfer from Expansion Opacity in Kilonova Ejecta

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ABSTRACT

Kilonova radiative-transfer codes that work from binned opacities describe their line treatment as "the Sobolev approximation"; what they apply is Sobolev's treatment of an isolated resonance together with the expansion-opacity closure built on it. These are two approximations, and they preserve different properties of a line forest. We measure each against frequency-resolved transfer in a deliberately restricted harness—a 1-D homologous shell, pure absorption with fixed LTE populations, controlled shell velocities of 1000–3000 $\mathrm{km\,s^{-1}}$—chosen because it makes the two errors separable. The expansion-opacity coefficient of Karp et al. and Eastman & Pinto is exact in the statistic it was derived for: the expected number of line interactions per unit path, $\sum_k (1 - e^{-\tau_k})$, is preserved identically. It departs from the deterministic attenuation of a resolved line realization because it applies Poisson statistics to what is a product of Bernoulli survivals, $e^{-\sum(1-e^{-\tau_k})}$ against $e^{-\sum \tau_k}$; the two coincide when every line is weak, and separate once individual lines saturate. On a forest of 153 calibrated La II lines the resulting band-flux difference is +38% at thermal line widths and at every transport bin width tried, growing from $\sim 3\%$ for weak-line forests to +38–48% for strong ones. The Sobolev localization itself, measured against a deterministic finite-profile reference on identical rays, carries no strength floor: its error is a boundary effect proportional to the ratio of Doppler width to the shell's velocity span, $\lesssim 2\%$ at $v_D \leq 30\,\mathrm{km\,s^{-1}}$ for the strengths relevant here and $\lesssim 0.3\%$ at $v_D \leq 10\,\mathrm{km\,s^{-1}}$. Across 36 further conditions spanning four wavelength windows, three epochs and three ion mixtures the separation is robust, and the controlling quantity is the transmission-weighted saturation deficit of the forest, for which the strongest line's optical depth is a proxy within a fixed geometry. The two approximations carried under one name fail for different reasons and at different levels. In one energy-conserving check with radiative equilibrium enabled, the band differential falls from +44% to between +5% and +8% as the removed flux is re-emitted redward.

Key words: radiative transfer – opacity – methods: numerical – stars: neutron – nuclear reactions, nucleosynthesis, abundances

1 INTRODUCTION

1.1 Kilonovae in three paragraphs

When two neutron stars spiral into each other and merge, a small fraction of their matter—between roughly a thousandth and a few hundredths of a solar mass—is flung outward at speeds of ten to thirty percent of the speed of light. This ejected matter is one of the very few environments in the universe dense enough in free neutrons to build the heaviest elements (gold, platinum, uranium, and the lanthanide row of the periodic table) through the rapid neutron-capture process, or *r-process*. The radioactive decay of these freshly made nuclei heats the expanding cloud, which then glows for days to weeks: this glow is a *kilonova* (for a review see Metzger 2020). One has been observed in detail, AT 2017gfo, the optical counterpart of the gravitational-wave event GW170817 (Abbott et al. 2017), whose modelling gave the first direct evidence that such mergers synthesize heavy r-process elements (Kasen et al. 2017).

Everything we want to learn from a kilonova—how much matter was ejected, how fast, and above all *which elements were made*—must be read out of its light. The light escapes only after filtering through the ejecta, and the filtering is dominated by an enormous number of atomic absorption lines. Lanthanide ions are the worst offenders: because of their complex open $4f$ electron shells, a single lanthanide ion can have tens of thousands of energy levels and millions of radiative transitions (Tanaka et al. 2020; Flörs et al. 2026b). Their presence raises the ejecta opacity by orders of magnitude and reddens and lengthens the transient (Barnes & Kasen 2013). The opacity of the ejecta, and therefore the color, brightness, and spectral features of the kilonova, is controlled by these line forests.

Simulating light transport through millions of lines, each intrinsically narrower than one part per million in frequency, is prohibitively expensive if done directly. Practical kilonova radiative-transfer codes therefore lean on an approximation

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introduced by Sobolev (1960) for expanding stellar envelopes. It reaches them in two distinct packagings. One family applies it line by line inside a Monte Carlo scheme with macroatom or downbranching redistribution (Lucy 2002, 2003; Kerzendorf & Sim 2014); the other bins it into the "expansion opacity" of Karp et al. (1977) and Eastman & Pinto (1993), which is what we test here. Section 2 explains both ideas from scratch. The distinction matters for how our results should be read, and we return to it in §6. The approximation is standard, venerable, and almost never checked against a resolved calculation *under kilonova conditions*. This work begins that check.

It also keeps separate two things that are often named together. Sobolev's approximation proper attenuates each line by $e^{-\tau_S}$ and rests on two assumptions: that the gas does not change across a resonance, and that lines do not overlap. Expansion opacity is a further, and separate, step—a statistical closure that converts each line's Sobolev interaction probability $1 - e^{-\tau}$ into a coarse-frequency coefficient. That the closure counts each line at most once was noted by Fontes et al. (2020); what we add is a measurement, on calibrated lanthanide data and against a resolved reference, of what that counting does to transmission and when it matters. A code that works from binned opacities and reports "the Sobolev approximation" is applying both steps, and the two carry error budgets differing by roughly an order of magnitude.

Several strands of previous work bear on this. Fontes et al. (2020) wrote the expansion-opacity optical depth as $\sum_i(1 - e^{-\tau_i})$ against the line-binned $\sum_i \tau_i$, noted explicitly that the former limits a single line's contribution to one, and compared both against a continuous Monte Carlo Sobolev calculation—which they call the more accurate treatment—in a one-dimensional pure-Nd test, finding broad agreement in the low-optical-depth regime that dominates their transport. Banerjee et al. (2022, 2024) showed that in the highly ionized early ejecta the line density can become large enough that intrinsic profiles overlap and the independent-interaction assumption underlying expansion opacity fails, particularly in the far ultraviolet. Separately, line-by-line Sobolev transport now follows tens of millions of transitions with fluorescence in multidimensional merger ejecta (Shingles et al. 2023; Collins et al. 2023, 2026), and Morag (2026) has shown that the Eastman & Pinto (1993) coefficient can capture photon mean free paths through a line forest while substantially underestimating emissivity and reprocessing rates.

Our question is complementary to each, and the novelty is specific. We neither propose a new opacity prescription nor construct a realistic kilonova spectrum. We hold the atomic data, populations, geometry and radiation source fixed and insert a finite-profile resolved calculation *ahead of* per-line Sobolev transfer—finite-profile transfer → per-line Sobolev transfer → binned expansion opacity—so that a measured discrepancy can be assigned to one step rather than to the stack; we map how each step's error depends on intrinsic width, line strength, bin width, line density and calibrated lanthanide line lists; and we identify the ensemble statistic that controls the closure's departure from resolved transmission. Appendix B sets out the relation to each of these treatments in more detail.

1.2 What this paper does

We do not build a kilonova pipeline. We build the smallest apparatus that can measure the two approximation errors cleanly and *separately*, and we operate it on progressively more realistic inputs. Concretely, this first paper reports:

- a **three-way validation harness** (Section 3): the same spherically expanding model, atomic data, and level populations evaluated by (a) analytic Sobolev theory, (b) a purpose-written deterministic frequency-resolved solver, and (c) the Monte Carlo code SEDONA (Kasen et al. 2006) in both its resolved and its expansion-opacity modes;
- **verification experiments** (Section 4) in which the resolved-physics calculations agree at the $\sim$ 1% level on the multi-line problems and, once the transport treatment is matched, on a single isolated line, establishing the baseline everything else is measured against;
- **the separation itself** (Section 5.3), the central result: on a forest of 153 calibrated La II lines from the GSI database (Flörs et al. 2026b,a), measured against a deterministic finite-profile reference on identical rays, the Sobolev localization errs by +3% at a $100\,\mathrm{km\,s^{-1}}$ Doppler width and by $\lesssim 0.3\%$ at $\leq 10\,\mathrm{km\,s^{-1}}$, while the expansion-opacity closure errs by +38% at all widths—and the two differ through different mechanisms;
- **the mechanism** (Sections 2.5, 5.3): the closure preserves the expected number of line interactions per crossing exactly and applies Poisson statistics to what is a product of Bernoulli survivals, so its departure from resolved transmission is the saturation deficit $\sum_k[\tau_k - (1 - e^{-\tau_k})]$ per ray; that statement is exact, and it is what makes the error per-resonance, width-independent, bin-width-invariant and non-monotonic in forest density;
- **validity maps** (Sections 5.4–5.6): both errors as functions of line strength, Doppler width, temperature, epoch, wavelength and composition—36 further conditions across which the separation is robust, with the transmission-weighted saturation deficit as the controlling statistic and the strongest line's optical depth as its proxy within a fixed geometry;
- a **bin-width control** (Section 5.5) showing the closure's departure to be invariant from 0.025 to 41 lines per transport bin, as the count-preserving property requires;
- **methodological results** (Table 3), among them an exact symmetry cancellation that makes gradient-based validity diagnostics conservative; a boundary law for the Sobolev localization error; and three conventions—thermal emission, spectral normalization and transport treatment—whose mismatch produces spurious few-percent discrepancies in exactly the cross-code comparisons such a study depends on, which is why the Sobolev differential is formed against a deterministic reference and the Monte Carlo code serves as its validation.

Readers with no radiative-transfer background should read Section 2 first; the appendix derives every formula used in the text at a level that assumes only basic calculus.

2 A COMPACT RADIATIVE-TRANSFER PRIMER

2.1 Light through matter: intensity, opacity, optical depth

Consider a beam of light of frequency ν traveling through a gas. Its strength is described by the *specific intensity* I_ν (energy per unit time, area, solid angle, and frequency). Along a path length s, the gas removes energy from the beam (absorption) and adds its own (emission):

$$\frac{\mathrm{d}I_\nu}{\mathrm{d}s} \;=\; -\,\alpha_\nu\, I_\nu \;+\; j_\nu. \tag{1}$$

Here α_ν (units cm^{-1}) is the *absorption coefficient*: $1/\alpha_\nu$ is the average distance a photon travels before being absorbed. The emissivity j_ν describes how much the gas glows. In this work the gas is kept in conditions where its own glow at the relevant frequencies is negligible or exactly modeled (Section 3.2), so absorption dominates.

The dimensionless quantity that controls absorption is the *optical depth*

$$\tau_\nu \;=\; \int \alpha_\nu\, \mathrm{d}s, \tag{2}$$

which counts how many absorption mean-free-paths the beam crosses. If the gas does not glow, the solution of Eq. (1) is simply

$$I_\nu^{\rm out} \;=\; I_\nu^{\rm in}\, e^{-\tau_\nu} : \tag{3}$$

a beam crossing optical depth $\tau = 1$ keeps 37% of its light, $\tau = 2$ keeps 14%, $\tau = 5$ keeps 0.7%. Every result in this paper is, at bottom, a statement about which τ gets exponentiated. (The full solution with emission is derived in Appendix A1.)

2.2 Spectral lines and Doppler width

Atoms absorb light most strongly at the discrete frequencies of their internal transitions—*spectral lines*. A line is not infinitely sharp: thermal motion of the absorbing ions Doppler-shifts each ion's absorption slightly, smearing the line into a profile $\phi(\Delta\nu)$ (normalized so $\int \phi\, \mathrm{d}\nu = 1$) whose width corresponds to a velocity

$$v_D \;=\; \sqrt{2k_BT/m_{\rm ion}}, \tag{4}$$

typically $\sim 0.6\,\mathrm{km\,s^{-1}}$ for lanthanum at kilonova temperatures. Expressed as a fraction of the line frequency this is $v_D/c \sim 2\times10^{-6}$: lines are extraordinarily narrow. The absorption coefficient of one line is

$$\alpha_\nu \;=\; \sigma_{\rm cl}\, f\, n_l\, \phi(\nu-\nu_0), \qquad \sigma_{\rm cl} \equiv \frac{\pi e^2}{m_e c} = 0.02654\ \mathrm{cm^2\,Hz}, \tag{5}$$

where ν_0 is the line's rest frequency, f its dimensionless *oscillator strength* (how intrinsically strong the transition is), and n_l the number density of ions sitting in the transition's lower energy level. How many ions sit in which level is set, in thermal equilibrium, by the Boltzmann distribution (Appendix A1).

2.3 Expanding ejecta: resonance surfaces

Kilonova ejecta expand *homologously*: after a short time every parcel coasts at constant speed, so its position is $r = v\,t$ and equivalently

$$v(r) \;=\; r/t, \tag{6}$$

where t is the time since the merger. Velocity increases linearly outward, with velocity proportional to radius.

Now follow a single photon of fixed frequency ν flying through this medium. Each gas parcel it passes sees the photon Doppler-shifted by the local line-of-sight velocity. Because the velocity changes monotonically along the flight path, the photon's frequency *as seen by the gas* sweeps continuously redward. It therefore comes into resonance with any given line at **one specific location** — the point where the local Doppler shift brings the gas-frame frequency onto the line center — and is out of resonance everywhere else. In homologous flow this location is particularly simple: for a photon traveling toward an observer along the z axis, the line-of-sight velocity of the gas at coordinate z is z/t for *every* ray, so the resonance locations are flat planes $z = \mathrm{const}$.

2.4 The Sobolev approximation

Sobolev (1960) realized that this geometry makes the line optical depth computable without resolving the profile at all; Castor (1970) cast the same idea in the escape-probability form used by most modern codes. The photon crosses the narrow resonance region quickly; if the gas properties (n_l, temperature) are constant across that thin region, the integral in Eq. (2) collapses (Appendix A2) to the *Sobolev optical depth*

$$\boxed{\tau_{\rm S} \;=\; \sigma_{\rm cl}\, f\, n_l(r_{\rm res})\, \lambda_0\, t} \tag{7}$$

where λ_0 is the line's rest wavelength and n_l is evaluated *at the resonance point*. Stimulated emission multiplies this by $(1-n_ug_l/n_lg_u)$, which for the LTE populations used throughout equals $1-e^{-h\nu/kT}$: 3×10^{-6} at 3800 Å and 3000 K, but 5×10^{-3} at 9100 Å. SEDONA applies it to every line; our analytic legs fold it into the level populations so that the two agree. Two assumptions are buried here and both are testable: (i) *locality* — the gas does not change across the resonance region (width $\sim v_D$ in velocity); (ii) *isolation* — no other line's resonance region overlaps this one. A photon crossing k line resonances is attenuated by $e^{-\sum_k \tau_{{\rm S},k}}$.

2.5 Expansion opacity

Codes that transport on a frequency grid need an opacity per frequency *bin*, not per line (line-by-line Monte Carlo schemes do not, and are discussed in Appendix B). The *expansion opacity* formalism (Karp et al. 1977; Eastman & Pinto 1993) assigns to each bin an effective absorption coefficient built from the Sobolev interaction probabilities of all the lines inside it,

$$\alpha_{\rm bin}^{\rm exp} \;=\; \frac{1}{c\,t} \sum_{k\in{\rm bin}} \frac{\nu_k}{\Delta\nu_{\rm bin}} \left(1-e^{-\tau_{{\rm S},k}}\right). \tag{8}$$

It is important to be precise about what this coefficient represents. Karp et al. (1977) derived it from the mean free path of a photon that redshifts continuously through a forest of lines: $1-e^{-\tau_k}$ is the probability that a photon crossing resonance k interacts there, so integrating $\alpha^{\rm exp}$ along the path on which the lines of one bin are swept gives $\sum_k(1-e^{-\tau_k})$—the *expected number of line interactions* per crossing for a photon that is counted but not removed. That expectation is what the closure is built to reproduce, and it reproduces it exactly (Appendix A4).

Transmission is a different statistic. A photon crossing k resonances is transmitted with probability $\prod_k e^{-\tau_k} = e^{-S}$, $S = \sum_k \tau_k$: a product of independent Bernoulli survivals. A transport code that treats $\alpha^{\rm exp}$ as an ordinary continuous absorption coefficient transmits e^{-E} with $E = \sum_k(1-e^{-\tau_k})$: the survival of a Poisson process with the same mean. The two agree whenever every $\tau_k \ll 1$, since then $1-e^{-\tau} \approx \tau$ and a sum of many small Bernoulli trials is Poisson; that is the many-weak-lines limit in which the formalism was derived and in which it is exact. They separate once individual lines saturate: for a single line with $\tau_{\rm S} = 2$ the transmission is 0.135 against

$$e^{-(1-e^{-\tau_{\rm S}})} = 0.421, \tag{9}$$

and for $\tau_{\rm S} = 50$ it is essentially zero against $e^{-1} = 0.37$. Per ray the difference is exactly the *saturation deficit*

$$D \;=\; S - E \;=\; \sum_k \left[\tau_k - (1-e^{-\tau_k})\right] \;\geq\; 0, \tag{10}$$

and for a band average over rays with weights w the identity $F_{\rm exp}/F_{\rm Sob} = \langle e^{-S}e^{D}\rangle_w / \langle e^{-S}\rangle_w$ holds without approximation (Appendix A4). **That the closure preserves the expected interaction count and not the transmission, and the size of the gap on calibrated lanthanide data, is the central object measured in this paper.**

This is the same distinction Morag (2026) draws from the emissivity side: the Eastman & Pinto (1993) prescription can capture photon mean free paths through a line forest while substantially underestimating photon emissivity and reprocessing rates. An effective opacity calibrated to reproduce one property of a line ensemble need not reproduce the others; here the preserved property is the interaction count and the unpreserved one is the transmission.

3 METHODS: THE THREE-WAY HARNESS

3.1 Design principle

The danger in comparing two radiative-transfer treatments is that differences can hide anywhere: different atomic data, different level populations, different geometry, different frequency grids. Following the design requirement that only the line-transfer treatment may differ, every experiment in this paper evaluates *one* physical configuration three ways:

Agreement of the first three (analytic, deterministic, SEDONA resolved) validates the harness; the offset of the fourth is the measured error, reported as

$$\Delta_{\rm exp} \;=\; \frac{F_{\rm exp} - F_{\rm resolved}}{F_{\rm resolved}}, \qquad \Delta_{\rm Sob} \;=\; \frac{F_{\rm Sob} - F_{\rm resolved}}{F_{\rm resolved}}, \tag{11}$$

where F is the band-averaged transmitted flux. The two are formed differently, and deliberately so. $\Delta_{\rm exp}$ is formed from the *two* SEDONA runs, so Monte Carlo conventions, frequency grids and normalizations cancel identically; it is also available analytically, from the same per-line machinery as the Sobolev leg with each crossing's τ replaced by $1-e^{-\tau}$, and the two agree at thermal widths (§5.3). $\Delta_{\rm Sob}$ is the finite-profile-versus-delta-resonance difference, and the cleanest quantity for it is a *deterministic* one: the Sobolev leg against the resolved formal solution of the same configuration on identical rays, with the same source convention, populations and transport treatment. We form it that way, using the closed-form resolved leg of Appendix A1 (Gaussian profiles, uniform populations, pure absorption), and use the SEDONA resolved run—at its seed mean—as the independent validation of that reference rather than as the reference itself. The reason is practical: a cross-code differential is exposed to three conventions that must be matched by hand—thermal emission (§3.6), spectral normalization (§5.6) and transport treatment (Appendix A7)—and each can produce a spurious few-percent result of the size being measured. Same-code differentials carry none of that exposure.

3.2 Geometry and physical conventions

All experiments use a spherical, homologously expanding shell (Eq. 6) around a central blackbody "light bulb" of radius $r_{\rm core}$ and temperature $T_{\rm core}$: the shell imprints absorption lines on the core's smooth continuum, and the emergent spectrum is measured far away. Rays that hit the core start with its blackbody intensity; rays that miss it start dark. Table 1 gives the numerical configuration in full.

Three deliberate simplifications keep the comparison interpretable:

(i) **Pure absorption, local thermal emission.** Scattering and fluorescence are off; absorbed energy is re-emitted thermally with source function $S_\nu = B_\nu(T)$ (the Planck function). Shell temperatures are chosen so the shell's own glow in the measured band is negligible ($B_\nu(T_{\rm shell}) \ll B_\nu(T_{\rm core})$); every leg treats the identical source, so any residual emission cancels in $\Delta_{\rm exp}$.

(ii) **Fixed temperatures.** Radiative equilibrium is disabled; the gas state never feedbacks on the radiation. Populations are therefore known exactly.

(iii) **Controlled velocities.** Shell velocities span 1000–3000 km s^{-1} rather than the realistic 0.1–0.3c. This keeps the difference between SEDONA's frozen steady-state transport and a worldline-consistent treatment negligible for the present measurements (F3/F11, Appendix A7), and keeps the frequency-grid cost of resolving line profiles affordable.

3.3 Population control

SEDONA computes its own thermal-equilibrium level populations from the atomic data file it is given. To guarantee both codes use *exactly* the same populations, we generate SEDONA's atomic-data files directly from the same source data the Python side reads: for the La II experiments, all 472 energy levels of the GSI La II model are written into the file, only the transitions in the studied wavelength window are included as lines, and the ionization potential is set to 10^5 eV so that ionization is impossible and the element sits entirely in the intended ionization stage. Both codes then evaluate the

Leg	Role
Analytic: Sobolev theory, Eq. (7), p-averaged over the source disk (App. A5)	prediction
Deterministic: **`formal_transfer.py`** (this work), frequency-resolved formal solution along rays	resolved reference
Monte Carlo: SEDONA (Kasen et al. 2006; Roth & Kasen 2015)	
(a) resolved bound–bound mode	resolved cross-check
(b) expansion-opacity mode	**under test**

Table 1. Numerical configuration of the production runs. The values are identical across every leg and every comparison unless a section states otherwise; the sweeps vary only $\tau_{\max}$ (through the ion density), v_D, $T_{\rm shell}$ and the epoch.

Quantity	Value
Core radius $r_{\rm core}$	8.64×10^{12} cm
Outer radius $r_{\rm out}$	2.592×10^{13} cm ($3\,r_{\rm core}$)
Epoch t	86400 s (day 1)
Velocity span $\Delta v_{\rm shell}$	1000–3000 km s^{-1}
Core temperature $T_{\rm core}$	6000 K
Core luminosity	6.894×10^{37} erg s^{-1}
Shell temperature $T_{\rm shell}$	3000 K (2500–5000 K on the T axis)
Fiducial Doppler width v_D	100 km s^{-1} (1–300 km s^{-1} swept)
Transport grid	$\nu \in [7.50, 7.95] \times 10^{14}$ Hz, $\Delta\nu/\nu = 4.17 \times 10^{-5}$
Spectrum grid	$\Delta\nu/\nu = 10^{-4}$
Band averaged	3800–3955 Å
Normalization margin	3952–3970 Å (line-free)
SEDONA packets	2×10^6 per run
Solver impact parameters	150 rays
Solver frequency points	1600
Electron scattering	off ($\epsilon = 1$)
Radiative equilibrium	disabled

same Boltzmann distribution over the same levels; we verified in SEDONA's output gas state that the free-electron fraction is $\sim 10^{-10}$, i.e. the population control holds.

3.4 The deterministic reference solver

The independent resolved leg is a deliberately small ($\sim$200-line) Python program. It casts parallel rays through the sphere at many impact parameters p, evaluates the frequency-resolved line opacity (Eq. 5) at every step using the local Doppler-shifted frequency, and accumulates the formal solution of Eq. (1) by cumulative optical depth (Appendix A1). Emergent luminosity is the p-weighted integral of ray intensities. Its correctness rests on four analytic limits enforced as unit tests: a bare core reproduces the exact blackbody-sphere luminosity; a single line's absorption trough reaches $e^{-\tau_{\rm S}}$ in the Sobolev limit; two well-separated lines absorb independently while overlapping ones multiply ($e^{-(\tau_1+\tau_2)}$); and a warm shell partially refills the trough. The full test suite of the accompanying repository (68 tests) passes at every commit reported here.

3.5 The Monte Carlo noise floor, measured

Every number quoted from SEDONA carries sampling noise, and since several results below are at the sub-percent level that noise has to be measured rather than asserted. We reran identical configurations with different random-number seeds, holding model, atomic data, grid and packet count fixed, so that the spread is pure sampling error. Five seeds at $(\tau_{\max}, v_D) = (5, 100\,{\rm km\,s^{-1}})$ give a fractional standard deviation of 0.26% on the resolved band flux and 0.30% on the expansion band flux; five seeds at $v_D = 10\,{\rm km\,s^{-1}}$ give 0.30%.

That the two widths agree to within 15% of each other matters. The band flux is a sum over all bins in the window, so its sampling error is set by the number of packets landing in the band rather than by how finely the band is subdivided, and should therefore be independent of v_D at fixed `core_n_emit`. Measuring at two widths an order of magnitude apart tests that expectation instead of assuming it, which is what licenses quoting the same uncertainty at the $v_D = 1\,{\rm km\,s^{-1}}$ frontier point, where repeating the run five times would cost some sixty times more.

We therefore attach $\pm 0.3\%$ (1σ) to every single-realization band flux. For $\Delta_{\rm exp}$ the two legs are not independent when they are run with the same fixed seed: the core-emission stream—packet positions, directions and frequencies—is then identical in both, and the streams diverge only at the first interaction. Measured on seed-matched pairs, the resolved and expansion band fluxes have a correlation coefficient of +0.97 and the paired $\Delta_{\rm exp}$ scatters by 0.12% (standard deviation over ten seeds; 0.04% on its mean), against 0.36% from adding the legs in quadrature. Every $\Delta_{\rm exp}$ quoted below is therefore a seed-matched pair—ten seeds for the headline forest, three at every point of the strength–width grid and the $3\,{\rm km\,s^{-1}}$ frontier—and every headline band flux is a seed mean with its standard error: for the La II forest the resolved mean is 0.3422 ± 0.0003 and the expansion mean 0.4958 ± 0.0004. The seed scatter of a single realization sits within $1.5\times$ the Poisson expectation from the packet count in the band at every grid point. Single realizations are quoted only where stated. (The solver-versus-SEDONA comparisons are the one place where a Monte Carlo reference still enters a cross-code difference; there the $\pm 0.3\%$ applies.)

3.6 A convention that must be matched: thermal emission

Finding F8, recorded because it is a trap for anyone benchmarking a deterministic solver against a Monte Carlo code. Our solver integrates the LTE source term $S_\nu = B_\nu(T_{\rm shell})$ along every ray; SEDONA, run at fixed temperature with radiative equilibrium disabled, deposits absorbed packet energy without re-emitting it. Setting $T_{\rm shell} \to 0$ in the solver isolates the difference exactly: the single-ion forest moves $0.3587 \to 0.3439$ against SEDONA's 0.3435 ($+4.4\% \to +0.1\%$), and the blend $0.2410 \to 0.2249$ against 0.2277 ($+5.8\% \to -1.2\%$). Like for like, the two independent codes agree to 0.1% on the single-ion forest—inside the 0.3% Monte Carlo noise of §3.5, so the residual is not resolvable—and to 1.2% on the blend. The term exceeds the naive $B_\nu(3000)/B_\nu(6000) = 2 \times 10^{-3}$ estimate because the shell subtends nine times the core's projected area, and it grows with saturation. Absolute band fluxes quoted from SEDONA in this configuration are therefore attenuation-only; $\Delta_{\rm exp}$, a differential between two SEDONA runs sharing one convention, is unaffected.

3.7 Atomic data

Real-line experiments use the GSI Database for Kilonova Radiative Transfer (Flörs et al. 2026b,a). It is one of several recent efforts to supply r-process opacity data (cf. Tanaka et al. 2020; Kato et al. 2024); we adopt it because its wavelengths are calibrated against experiment, which matters when the physical question turns on separations of a few $\mathrm{km\,s^{-1}}$. It provides calibrated energy levels and E1 transitions for singly and doubly ionized lanthanides, with per-level calibration flags (`xmatch` = matched to experiment, `shifted` = calibrated shift, `uncalib`). For La II: 472 levels and 17,743 transitions. Internal consistency of the dataset is excellent: oscillator strengths recomputed from the Einstein A coefficients agree with the tabulated $\log gf$ values to 1.5×10^{-4} across our window, so both codes carry identical line strengths regardless of which column they read.

Atomic-data uncertainty is itself a leading error source in kilonova modelling, and it is deliberately *not* varied here. Improved calculations for singly ionized lanthanides change individual-element opacities by factors of several and mixture opacities by tens of per cent (Kato et al. 2024), and a recent comparison of three neodymium datasets found that changing the input line list alone moves kilonova spectra substantially and the peak bolometric luminosity by close to a factor of 1.5 (Fontes et al. 2026). The present experiment addresses the orthogonal budget: every transport leg receives byte-identical atomic data and populations, so the differences reported here measure the transport approximation rather than uncertainty in the line list. They should be read as additional to the atomic-physics uncertainty now being quantified, not as an estimate of it.

4 VERIFICATION EXPERIMENTS

4.1 Reproducing and breaking the Sobolev limit in isolation

Before any code comparison, the toy problem: one line, constant n_l, homologous slab. The resolved integral (Eq. 2) converges to the analytic $\tau_{\rm S}$ (Eq. 7) to machine precision ($E_{\rm Sob} = 2 \times 10^{-16}$ at 2×10^5 grid points), with monotone convergence under refinement—important because an unconverged grid mimics a physical Sobolev deviation.

Breaking locality on purpose (Fig. 1) produced **Finding F1**: with the population gradient placed *symmetrically* about the resonance point, the error cancels identically—the line profile is even while the leading gradient term is odd, and the natural symmetric choice (tanh centered on resonance) also has vanishing curvature there. The error only appears when the resonance sits off-center on the gradient (we use the point where curvature is large), and then follows a clean ϵ^2 law, at the few-percent level when $\epsilon = v_D/v_{\rm scale} \gtrsim 0.3$ (Appendix A3). The practical consequence: widely used validity diagnostics based on the gradient magnitude $G = v_D\,|{\rm d}\ln n_l/{\rm d}v|$ are *conservative*: the true error depends on where the resonance sits on the gradient, and can be far below the G-based bound.

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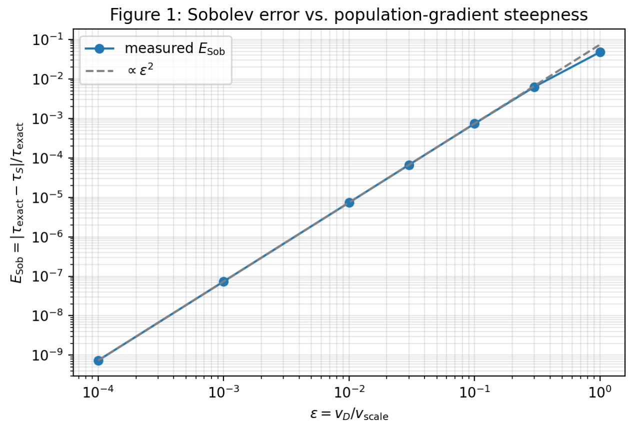

Figure 1. Controlled breakdown of the Sobolev locality assumption. A tanh-shaped population gradient of velocity scale $v_{\rm scale}$ is imposed; the relative error $E_{\rm Sob}$ between the resolved integral and the Sobolev formula follows the predicted ϵ^2 power law in $\epsilon = v_D/v_{\rm scale}$ (measured log-slope 2.0000), reaching $\sim 5\%$ when the gradient scale approaches the Doppler width.

analytic $e^{-\tau_{\rm S}}$ ($\beta \to 0$)	0.1353
frozen-snapshot law, $e^{-\tau_{\rm S}(1-\beta)/\gamma}$ over the trough	0.1371
deterministic solver, frozen mode	0.1372
SEDONA resolved, production grid	0.1420
SEDONA resolved, converged (5 seeds, 3.2×10^7 packets)	0.1383 ± 0.0001
SEDONA expansion opacity	**0.4285**
Poisson value $e^{-(1-e^{-\tau_{\rm S}})}$	0.4212

4.2 One line, three codes

The minimal cross-code experiment uses SEDONA's built-in two-level test atom (one line at 1215.5 Å, $f = 0.665$) in a shell tuned to $\tau_{\rm S} = 2$. The shell temperature (2000 K) was chosen after an ionization- balance check: at the very low densities that give $\tau_{\rm S} \sim 1$, a 5000 K gas would be fully ionized, destroying population control; at 2000 K the gas is neutral to one part in 10^{10}.

The measured absorption-trough depths (Fig. 2):

Two things have to be said about the resolved rows. First, the target for a code that iterates on a fixed snapshot of the ejecta is not $e^{-\tau_{\rm S}}$ but the frozen-snapshot law of Appendix A7, $e^{-\tau_{\rm S}(1-\beta)/\gamma}$, which over the 1400–2600 $\mathrm{km\,s^{-1}}$ trough window is 0.1371; the deterministic solver in frozen mode reproduces it to 0.1% under every variation of ray count, shell emission and profile truncation tried, so the 1.4% between it and 0.1353 is the transport treatment, not an error. Second, SEDONA's production run sits 3.5% above that target. Its transport grid there had ${\rm d}\nu/\nu = 2 \times 10^{-4}$, under two bins per Doppler width, against the eight used in every forest run, and the bin-width study of §5.5 shows the

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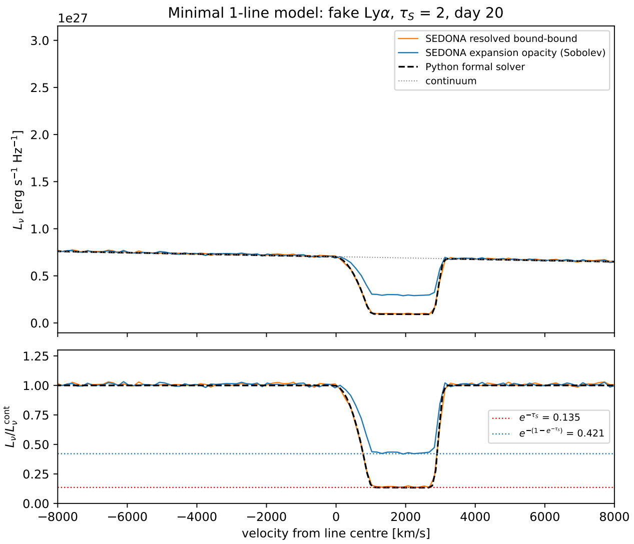

Figure 2. Minimal one-line model (artificial Lyα-like line, $\tau_\mathrm{S}=2$, cold neutral shell). Top: emergent spectra. Bottom: ratio to the continuum. SEDONA's resolved mode (orange) and the deterministic solver (black dashed) follow the same profile; the red dotted line is the analytic $e^{-\tau_\mathrm{S}}=0.135$, the $\beta\to 0$ limit of the frozen-snapshot law both codes solve (Appendix A7). The expansion-opacity mode (blue) sits at the Poisson value $e^{-(1-e^{-\tau_\mathrm{S}})}=0.421$ (blue dotted).

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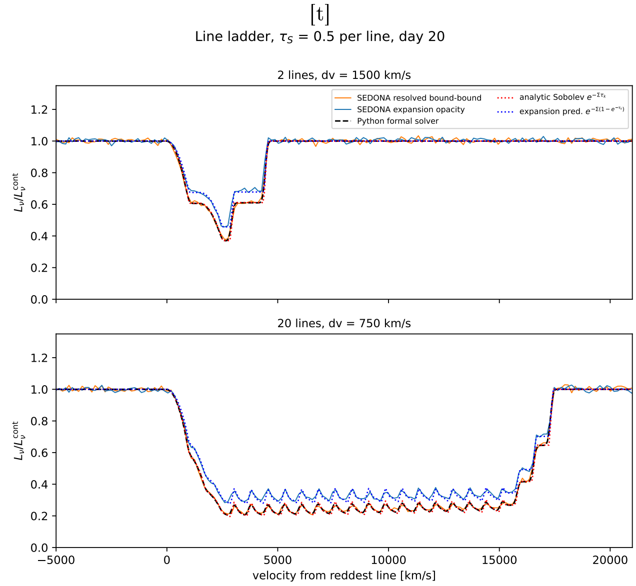

Figure 3. Line ladder at $\tau_\mathrm{S}=0.5$ per line. Top: two lines separated by $1500\,\mathrm{km\,s^{-1}}$ (overlapping troughs). Bottom: twenty lines spaced $750\,\mathrm{km\,s^{-1}}$ (a mini forest). The resolved trio—SEDONA resolved (orange), deterministic solver (black dashed), p-averaged analytic staircase (red dotted)—lock together tooth for tooth; the expansion pair (blue solid: SEDONA; blue dotted: analytic cap prediction) ride systematically above them.

Table 2. Single-line benchmark: SEDONA resolved trough depth (1400–$2600\,\mathrm{km\,s^{-1}}$ window) along one axis at a time, fixed seeds, three seeds per rung (five at the anchor). The frozen-snapshot target is 0.1371; the production run sat at 0.1420 with 2×10^6 packets on a 2×10^{-4} grid.

axis	value	trough	vs frozen
packets ($\mathrm{d}\nu/\nu = 2\times10^{-4}$)	5×10^5	0.1423 ± 0.0031	$+3.8\%$
	2×10^6	0.1406 ± 0.0017	$+2.6\%$
	8×10^6	0.1390 ± 0.0005	$+1.4\%$
	3.2×10^7	0.1386 ± 0.0001	$+1.1\%$
transport $\mathrm{d}\nu/\nu$ (8×10^6)	4×10^{-4}	0.1382 ± 0.0003	$+0.8\%$
	2×10^{-4}	0.1390 ± 0.0005	$+1.4\%$
	1×10^{-4}	0.1388 ± 0.0004	$+1.2\%$
	4.2×10^{-5}	0.1385 ± 0.0004	$+1.0\%$
	2×10^{-5}	0.1381 ± 0.0002	$+0.7\%$
spectrum $\mathrm{d}\nu/\nu$	5×10^{-4}	0.1385 ± 0.0004	$+1.0\%$
	2×10^{-4}	0.1384 ± 0.0005	$+0.9\%$
	1×10^{-4}	0.1385 ± 0.0004	$+1.0\%$
zones	25	0.1383 ± 0.0006	$+0.8\%$
	100	0.1384 ± 0.0005	$+0.9\%$
	101 (half-zone shift)	0.1387 ± 0.0002	$+1.2\%$
	400	0.1379 ± 0.0003	$+0.6\%$
anchor (3.2×10^7, 4.2×10^{-5}, 5 seeds)	resolved	0.1383 ± 0.0001	$+0.9\%$
	expansion	0.4438 ± 0.0002	(Poisson value 0.4212)
zero opacity	$\rho\times10^{-6}$	1.0079	–

resolved leg degrading exactly once bins exceed the profile. A convergence ladder in packet number, transport and spectrum grids, zone count and line-to-cell phase, with fixed seeds (Table 2), closes the gap to 0.1383 ± 0.0001 at the converged grid—a residual $+0.9\%$ above the frozen target that does not move with packet count, grid or zoning. The zero-opacity control identifies it: with the line removed the same trough window reads 1.0079 ± 0.002 against the red-margin normalization, i.e. SEDONA's continuum is 0.8% high there relative to the analytic Planck shape (its core packets are emitted in the comoving frame of the core surface and boosted to the lab frame). Dividing it out gives 0.1372, the frozen target to 0.1%. The production value 0.1420 was a 1.5σ draw at 2×10^6 packets (0.1406 ± 0.0017 over three seeds at that count). The expansion-opacity mode lands at 0.4438 ± 0.0003 at the converged grid (0.440 after the same continuum correction), 4.5% above the Poisson value 0.4212—the expansion-leg bin systematic of §5.5 on a single line—and a factor 3.2 above the resolved trough: the single-line statement of the count-versus-transmission distinction of §2.5, and the property Fontes et al. (2020) describe as limiting each line's contribution to one.

The same experiment surfaced the transport-treatment convention itself: the solver's excess over $e^{-\tau_\mathrm{S}}$ scales with the bulk velocity at the resonance, reaching 7% at $\beta=0.077$ and matching $e^{-\tau_\mathrm{S}(1-\beta)}$ to four digits. Appendix A7 shows this is the law of a *frozen snapshot* of the ejecta, which is what a steady-iterate transport calculation solves; letting the ejecta age advance along the photon path, as it physically must, gives instead $\tau/\tau_\mathrm{S}=1/\gamma$, with no first-order term, and the two treatments must be matched in any cross-code comparison. At the shell velocities used here the difference is $\lesssim 1\%$ in τ and is held fixed by running the solver in the mode that matches the code it is compared with.

4.3 From two lines to a twenty-line ladder

Custom SEDONA atomic files with 2 and then 20 identical-strength lines ($\tau_\mathrm{S}=0.5$ each) test multi-line behavior

analytic Sobolev (p-averaged)	0.2405
deterministic solver	0.2407
SEDONA resolved	0.2435
analytic expansion cap	0.3194
SEDONA expansion opacity	0.3257

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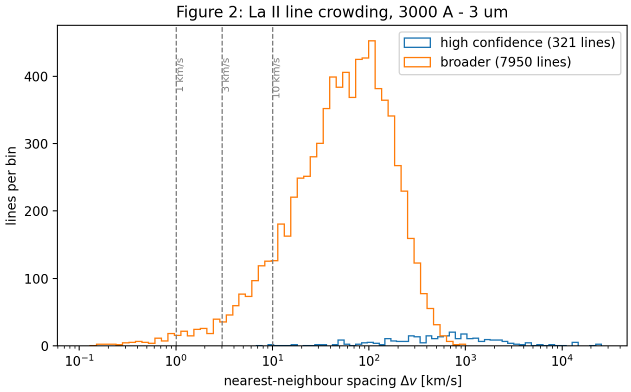

Figure 4. Nearest-neighbor velocity spacing of GSI La II lines (3000 Å–3 μm): in the broader subset, 11.3% of lines have a neighbor within 10 km s^{-1} and 3.3% within 3 km s^{-1}.

(Fig. 3). In the center of the 20-line forest, where each photon crosses ~ 3 resonances, five determinations of the transmitted fraction give:

deterministic solver, $S_\nu = B_\nu(T_{\rm shell})$	0.3587
deterministic solver, $T_{\rm shell} \to 0$	0.3439
SEDONA resolved	0.3435
SEDONA expansion opacity	0.4973
$\Delta_{\rm exp}$	**+44.8%**

Finding F5: the expansion-opacity error tracks its own per-line cap prediction, i.e. the error is incurred *per resonance crossing*. Adding more lines multiplies both the true and the capped attenuations equally, so the fractional flux error does *not* average away with line count—it only vanishes as $\tau \to 0$ per line. Expansion opacity is a *weak-line* approximation, not a many-line one.

A technical note that matters for anyone reproducing this: the analytic staircase must be averaged over the stellar disk (p-average, Appendix A5), because resonance planes closer than the core radius are crossed by only the outer rays; naive plane counting predicts visibly too-shallow troughs.

5 RESULTS: TWO APPROXIMATIONS, TWO ERRORS

With the harness validated, we turn to calibrated La II data. We first ask whether the assumptions underlying either approximation have anything to fear in real line lists (§5.1), then set up a realistic forest (§5.2) and measure both errors on it (§5.3)—the paper's central result. The remaining subsections establish where each error lives (§5.4), that the picture is not peculiar to one window or epoch (§5.6), how blending changes it (§5.7), and what the shortcut buys in computer time (§5.8).

5.1 Do the assumptions have anything to fear? Line crowding

The geometric isolated-line condition becomes questionable when neighbouring profiles lie within a Doppler width. That regime is not hypothetical in r-process ejecta: Banerjee et al. (2022, 2024) defined a critical opacity at which the characteristic line spacing becomes comparable to the intrinsic thermal width, and found that highly ionized lanthanides exceed it in the early-time far ultraviolet, so that the independent-interaction assumption entering expansion opacity can fail there. Whether crowding produces a *transport* error depends on the interaction physics, and §5.3 shows that it does not in the pure-absorption case studied here, where optical depths remain exactly additive. Our line-spacing diagnostic therefore asks a narrower question: whether analogous crowding exists in the lower-ionization data used here. Fig. 4 shows that even for this *single* ion the calibrated data contain a real crowding tail: 11.3% of the broader line sample has a neighbor within 10 km s^{-1}. Multi-ion mixtures only add lines, so the geometrically crowded regime exists in nature—which is what makes it worth establishing that crowding alone is not sufficient for a transport error.

5.2 A realistic forest: 153 calibrated lines

We select the 100 Å window with the richest optical-depth distribution (3850–3950 Å), build the population-controlled SEDONA atom from the GSI data (Section 3.3), and run the three-way comparison at $T = 3000$ K, $t = 1$ day, Doppler width 100 km s^{-1} (Fig. 5). Band-averaged transmitted flux over 3800–3955 Å:

The resolved pair agree to 4.4% as tabulated—and to within 1% once the thermal-emission convention is matched (Finding F8, §3.6) and the comparison is made at the seed mean rather than on a single realization—while the expansion-opacity run overestimates the transmitted band flux by 45%. Notably this exceeds the uniform-ladder difference (+33.8% at $\tau = 0.5$ per line): real forests are dominated by a few strong lines, precisely where the Poisson survival departs most from the Bernoulli product. That observation prompts the question the next subsection answers—how much of the difference belongs to the Sobolev localization and how much to the closure built on it.

5.3 Separating the two errors

We now put both approximations to the same forest and the same truth. The distinction drawn in §2.5 becomes operational here (Fig. 7): the Sobolev approximation *itself* attenuates each line by $e^{-\tau_S}$ with delta-function resonances, while expansion opacity substitutes $1 - e^{-\tau_S}$ at every crossing. In this pure-absorption LTE configuration the Sobolev prediction is exact analytics (the p-averaged staircase of Appendix A5), so it requires no additional transport calculations, and the identical machinery yields the expansion prediction by damping each crossing. A Monte Carlo realization of per-line Sobolev interactions would add only sampling noise to a result available in closed form.

For the La II forest at $v_D = 100$ km s^{-1}, against the deterministic finite-profile reference on identical rays ($F^{\rm det}_{\rm res} =$

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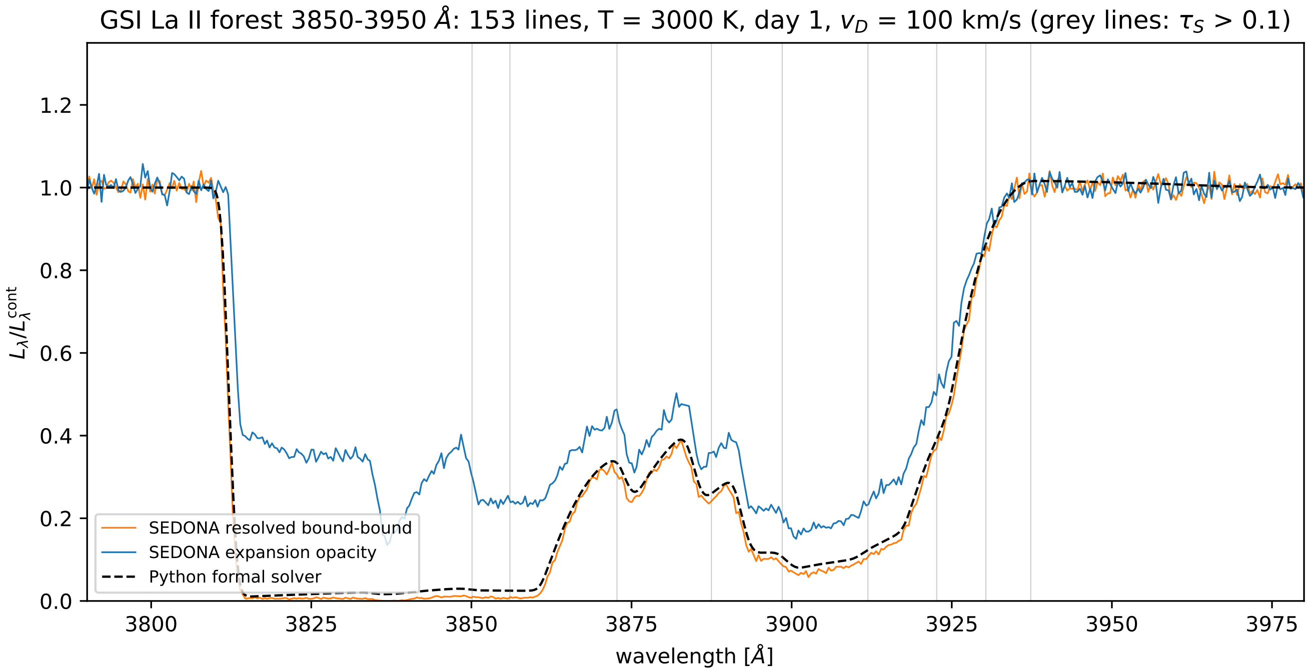

Figure 5. The 3850–3950 Å La II forest (153 calibrated lines, 3 with $\tau_{\rm S} > 1$, max 5) at $T = 3000$ K, one day after merger. The resolved treatments (orange, black dashed) saturate to black where strong troughs overlap; expansion opacity (blue) never drops below ~ 0.2 because each crossing is capped.

$\tau_{\rm max}$	v_D [km s^{-1}]	$\Delta_{\rm Sob}$	$\Delta_{\rm exp}$	$\Delta_{\rm exp}^{\rm analytic}$
0.5	10	+0.01%	+2.9%	+2.9%
0.5	300	+0.35%	+3.4%	+3.2%
5	10	+0.30%	+38.9%	+38.4%
5	300	+9.6%	+53.4%	+51.3%
50	10	+0.82%	+45.0%	+44.1%
50	300	+32.7%	+93.9%	+89.6%

v_D [km s^{-1}]	$\Delta_{\rm Sob}$	$\Delta_{\rm exp}$ (SEDONA)	$\Delta_{\rm exp}^{\rm analytic}$
1	+0.03%	+39.1%	+38.1%
3	+0.09%	+38.6%	+38.1%
10	+0.30%	+38.9%	+38.4%
30	+0.91%	+40.4%	+39.3%
100	+3.09%	+44.9%	+42.3%
300	+9.64%	+53.4%	+51.3%

0.3405): Sobolev proper gives 0.3511 (+3.1%) while the expansion-opacity closure gives 0.4848 (+42.4%). The SEDONA resolved seed mean over 10 matched seeds, 0.3422 ± 0.0003, validates the reference to 0.5%; against it the Sobolev leg is +2.6%, and SEDONA's own expansion run (0.4958 ± 0.0004) gives a paired $\Delta_{\rm exp}$ of +44.9%. Across the sweep, every $\Delta_{\rm Sob}$ below is formed against the deterministic reference and every $\Delta_{\rm exp}$ is a seed-matched SEDONA pair (Fig. 6):

The last column is the closure evaluated analytically—the same per-line machinery as the Sobolev leg with each crossing's τ replaced by $1 - e^{-\tau}$, i.e. $\langle e^{-E} \rangle$ of §2.5. It agrees with SEDONA's expansion run at narrow widths and sits 2–5 points below it at $v_D = 100$–300 km s^{-1}; the difference is SEDONA's own binned implementation, which §5.5 shows to carry a $\sim 2\%$ bin-width systematic in the expansion leg at fixed reference. We quote both, because the analytic value is the pure count-versus-transmission number and the SEDONA value is what a code actually delivers.

Down the width axis at $\tau_{\rm max} = 5$—free for the deterministic reference, whose cost does not depend on v_D—the Sobolev error simply switches off while the closure's does not:

Finding F9: the two components of F6 have different owners. (F6, from §5.4 below, decomposes Δ into a strength-set floor plus a term that grows with v_D.) *The strength-set floor belongs to the expansion-opacity closure alone.* The Sobolev localization exhibits no strength floor—at $v_D = 10$ km s^{-1} it is +0.01%, +0.30% and +0.82% at $\tau_{\rm max} = 0.5$, 5 and 50, following the boundary law of Appendix A6 rather than any saturation—while the closure climbs +3% → +38% → +45% across the same three. That floor is the Poisson-versus-Bernoulli gap of §2.5, not a failure of locality or isolation. *The width-growing term, by contrast, is not a strength effect at all*: a delta-function resonance has no width by construction, so the Sobolev leg is exactly v_D-independent, and the resolved calculation's width dependence appears as clipping of line profiles against the boundaries of the finite line-forming region (Appendix A6), proportional to $v_D/\Delta v_{\rm shell}$ and to the strength of the lines near the edges. It reaches +32.7% at $(\tau_{\rm max}, v_D) = (50, 300\,{\rm km\,s^{-1}})$—where v_D is 15% of the shell's velocity span, a configuration with no counterpart in real ejecta, for which $v_D/\Delta v_{\rm shell} \sim 10^{-4}$.

The headline is therefore sharper than either treatment alone suggests: at kilonova-relevant line widths **the Sobolev localization is accurate to $\lesssim 1\%$ at $v_D \leq 30$ km s^{-1}**

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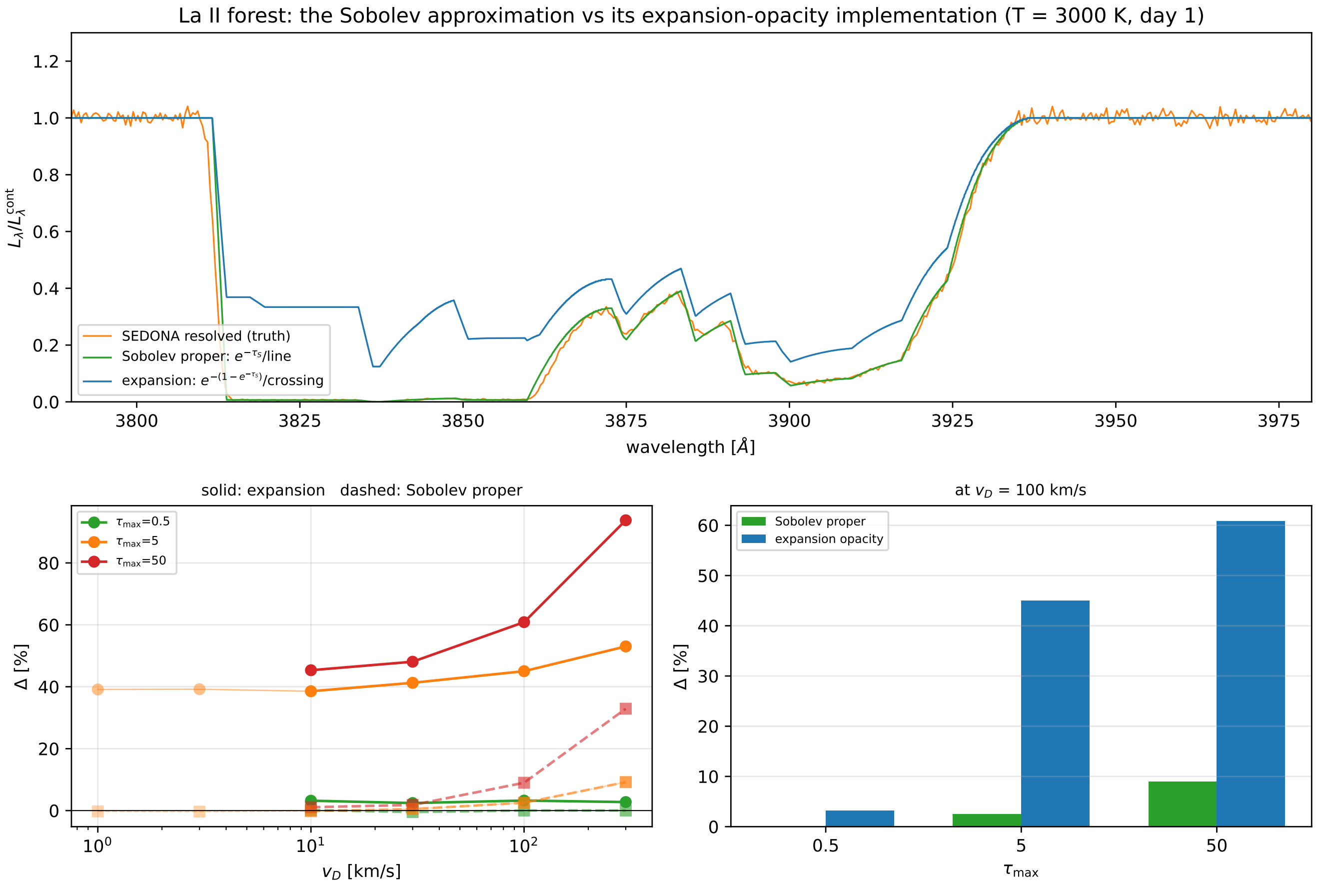

Figure 6. Top: the La II forest under three treatments. The Sobolev approximation proper (green, per-line $e^{-\tau_S}$) tracks the resolved calculation (orange) closely; expansion opacity (blue) sits well above it. Bottom left: both errors versus Doppler width, for three line-strength scalings (solid: expansion; dashed: Sobolev). Bottom right: the two at $v_D = 100\,\mathrm{km\,s^{-1}}$.

for $\tau_{\max} \leq 5$ (2.5% at $\tau_{\max} = 50$), and to $\lesssim 0.3\%$ at $v_D \leq 10\,\mathrm{km\,s^{-1}}$ across the full strength range, while the expansion-opacity closure differs from resolved transmission by +38–45% at every width. Two approximations are carried under one name; they differ through different mechanisms and at different levels.

5.3.0.1 Overlap is not the residual, and cannot be.

It is tempting to attribute the remaining Sobolev error at broad v_D to the isolation assumption, given the line crowding documented in §5.1. In this harness that is impossible. With fixed populations and pure absorption the opacities add, so $\tau_\nu = \sum_i \int \alpha_{\nu,i}\,\mathrm{d}s = \sum_i \tau_{\nu,i}$ *exactly*, and the emergent intensity is $I_\nu = I_{\nu,0}\exp(-\sum_i \tau_{S,i})$ whether or not the profiles overlap. Two identical lines driven from 20 Doppler widths apart to exact coincidence reproduce the Sobolev prediction to six decimal places. Line overlap becomes a genuine effect only once scattering, fluorescence or NLTE populations let one transition modify the radiation another one sees—all switched off here by design. The isolation assumption simply cannot be probed by an attenuation-only calculation.

The residual is instead *geometric*, and Appendix A6 derives it: where a resonance lies within a few Doppler widths of a boundary of the line-forming region, the resolved profile is clipped while Sobolev applies a hard step, giving $\tau/\tau_S = \frac{1}{2}[\mathrm{erf}((z_{\rm hi} - z_{\rm res})/\Delta) - \mathrm{erf}((z_{\rm lo} - z_{\rm res})/\ell_S)]$ with the Sobolev length $\ell_S = v_D t$. The affected fraction of the band is $\sim v_D/\Delta v_{\rm shell}$, which is why the error is invisible at thermal widths and reaches tens of percent only when v_D is a sizeable fraction of the velocity span of the shell.

5.4 Where each error lives: strength, width, temperature

The separation just established is a single point in parameter space. We now map where each error lives, by scaling the two quantities the two approximations should respond to differently: the overall line strength, which drives the per-crossing cap, and the Doppler width, which the Sobolev treatment ignores by construction.

Scaling the overall line strength ($\tau_{\max} \in \{0.5, 5, 50\}$ by scaling the ion density) and the Doppler width ($v_D \in \{1, 3, 10, 30, 100, 300\}\,\mathrm{km\,s^{-1}}$) yields the first rows of the validity map (Figs. 8–9; $\Delta_{\rm exp}$ in percent):

Finding F6: the error decomposes into

(i) a *strength-set floor*, independent of v_D: the pure per-

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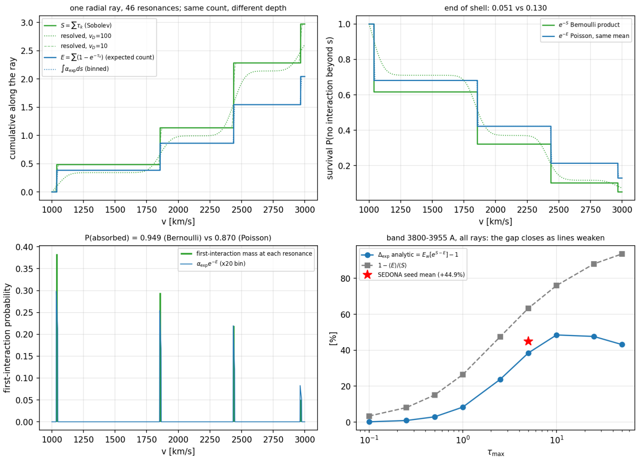

Figure 7. Count versus transmission along one radial ray through the La II forest (46 resonances). Top left: cumulative Sobolev depth S (green stair), its resolved counterpart at $v_D = 100$ and $10\,\mathrm{km\,s^{-1}}$ (dotted, dashed), the expected count E (blue stair) and the binned $\int \alpha_{\rm exp}\, {\rm d}s$ (dotted), which reaches the same E at the far edge. Top right: survivals $e^{-S} = 0.051$ and $e^{-E} = 0.130$ at the edge. Bottom left: the first-interaction distribution, discrete masses at the resonances against the smooth $\alpha_{\rm exp} e^{-E}$ (0.949 against 0.870 integrated: the closure under-counts absorptions). Bottom right: band-integrated $\Delta_{\rm exp}$ against $\tau_{\max}$ with $1 - \langle E\rangle/\langle S\rangle$ and the SEDONA seed mean. A 2×10^6-photon Monte Carlo on the same ray reproduces every panel to its sampling error.

$\tau_{\max}$	v_D [km s^{-1}]					
	1	3	10	30	100	300
0.5	–	–	+2.9	+3.0	+3.3	+3.4
5	+39.1	+38.6	+38.9	+40.4	+44.9	+53.4
50	–	–	+45.0	+47.5	+60.3	+93.9

crossing substitution error. Weak forests are safe at the 3% level; by $\tau_{\max} = 5$ the floor is $\sim 39\%$;

(ii) a *wing term* growing with v_D: expansion opacity is width-blind (its transmitted flux is flat across a factor 300 in v_D), while genuinely wide lines absorb continuum between resonances.

The frontier measurement at $1\,\mathrm{km\,s^{-1}}$—within a factor 2 of the actual La thermal width—confirms that the floor survives: *no refinement of line widths rehabilitates expansion opacity in the strong-line regime.*

The temperature axis (Fig. 9, left), with strength pinned, moves $\Delta_{\rm exp}$ only from +41.8% to +48.1% across 2500–5000 K: in this window the same few strong lines dominate at all temperatures, so population redistribution is a weak axis (single-ion result; multi-ion mixtures may differ).

5.5 Is the error an artifact of bin choice?

Expansion opacity is derived for frequency bins containing *many* lines (Karp et al. 1977; Eastman & Pinto 1993); the statistical averaging is what gives the construction its meaning. Our transport grids are much finer than that—$12.5\,\mathrm{km\,s^{-1}}$ bins against a mean La II line spacing of $\sim 50\,\mathrm{km\,s^{-1}}$, i.e.

line list	lines/bin	$F_{\rm resolved}$ spread	$\Delta_{\rm expansion}$
La II (153 lines)	0.025–2.5	0.13%	$+43.7\% \pm 1.2$
+ Ce II (2529)	4.1–41	0.5%	$+15.3\% \pm 0.7$

	median	max	$\tau_{\max} > 3$	$\tau_{\max} < 0.5$
$\Delta_{\rm exp}$ (SEDONA pair)	+5.1%	**+62.1%**	+20.1%	+0.3%
$\Delta_{\rm Sob}$ (deterministic ref.)	+0.26%	**+7.6%**	+2.57%	+0.01%
$\Delta_{\rm Sob}$ (SEDONA ref.)	+0.34%	+7.8%	+2.69%	−0.12%

about one line per four bins—so it is fair to ask whether we have measured the formalism outside its design regime. The question also bears on how our result sits beside Fontes et al. (2020), who compared line-binned, expansion-opacity and continuous-Sobolev calculations for a one-dimensional Nd model and obtained broadly similar spectra. Our test is narrower: rather than comparing complete synthetic spectra, it varies the transport bin width at fixed underlying physics, and asks whether the attenuation offset survives on entering the many-lines-per-bin regime.

Appendix A4 answers this analytically: the bin width cancels, so the expected interaction count $\sum_k (1 - e^{-\tau_k})$ is preserved at any resolution, and with it the Poisson survival that differs from the Bernoulli product. The numerical test confirms it. Sweeping the transport resolution over two decades at fixed physics (Fig. 10):

The two line lists give different values—the blend is denser and partly blanketed, which is F7—but neither depends on the bin width, including at 41 lines per bin, which is squarely inside the many-line regime the construction targets. The departure is a property of the count-preserving closure, not of how finely it is binned. Within the converged range the expansion leg itself does move by $\sim 2\%$ with bin width at fixed reference (0.487, 0.498, 0.487 at 1.25, 12.5 and $125\,\mathrm{km\,s^{-1}}$ bins), a systematic of SEDONA's binned implementation that the analytic closure does not have; it is the reason the SEDONA and analytic $\Delta_{\rm exp}$ columns of §5.3 differ by a few points at $v_D \geq 100\,\mathrm{km\,s^{-1}}$.

The sweep is restricted to bins narrower than $1.5\,v_D$ because the *resolved* reference must itself be converged: past that point it is no longer resolving anything, its band flux climbs from 0.342 to 0.679, and the comparison measures the reference rather than the approximation. Any verdict on the closure's bin-width dependence is conditional on that check.

5.6 A breadth sweep: windows, epochs and ion mixtures

The measurements above rest on one wavelength window at one epoch. To test their generality we ran a sweep over four windows (4300, 4900, 7000 and 9100 Å, selected automatically by optical-depth richness and deliberately excluding the window used so far), three epochs ($t = 0.5$, 1 and 3 days at fixed ejecta mass, so $\rho \propto t^{-3}$ and $\tau_{\rm S} \propto t^{-2}$), and three ion mixtures (La II; +Ce II; +Ce III). That is 36 conditions and 72 transport calculations, spanning 71 to 2504 lines per window and realized $\tau_{\max}$ from 8×10^{-5} to 34.

Finding F10: the separation is robust across the axes tested, and within this slice the realized $\tau_{\max}$ is a good empirical proxy for the controlling statistic. All 36 conditions follow one trend (Fig. 11) regardless of window, epoch or composi-

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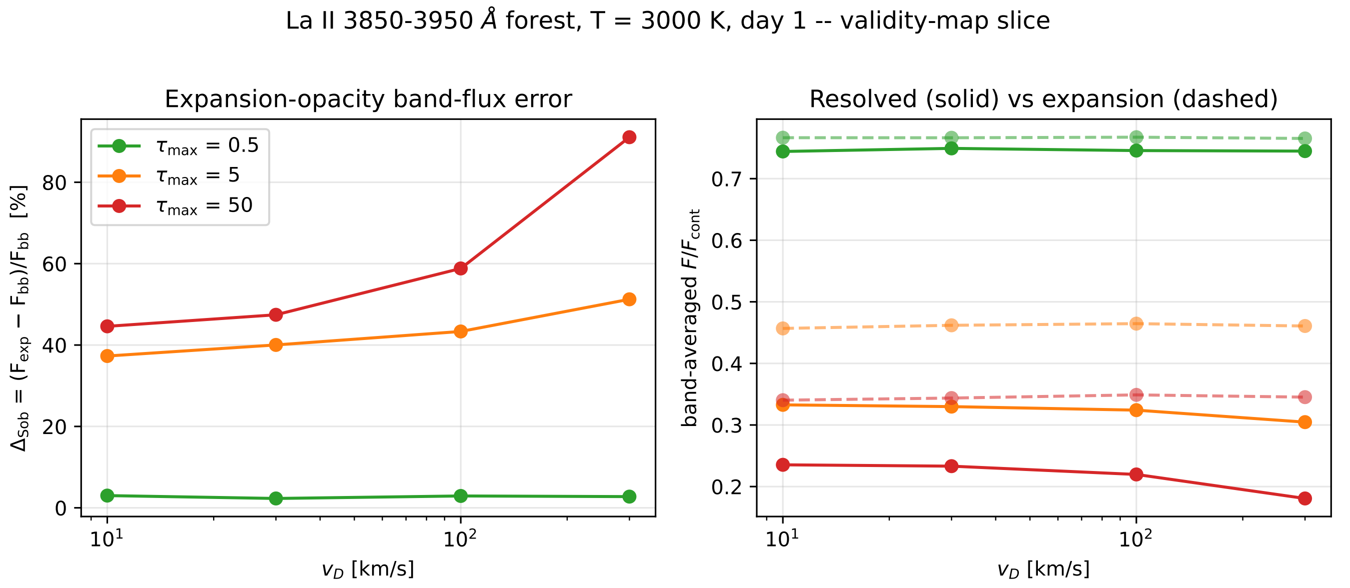

Figure 8. First validity-map slice: Δ_{exp} vs Doppler width for three overall line-strength scalings of the La II forest. Right panel: expansion-opacity fluxes (dashed) are blind to v_D, while resolved fluxes (solid) drop as profiles widen.

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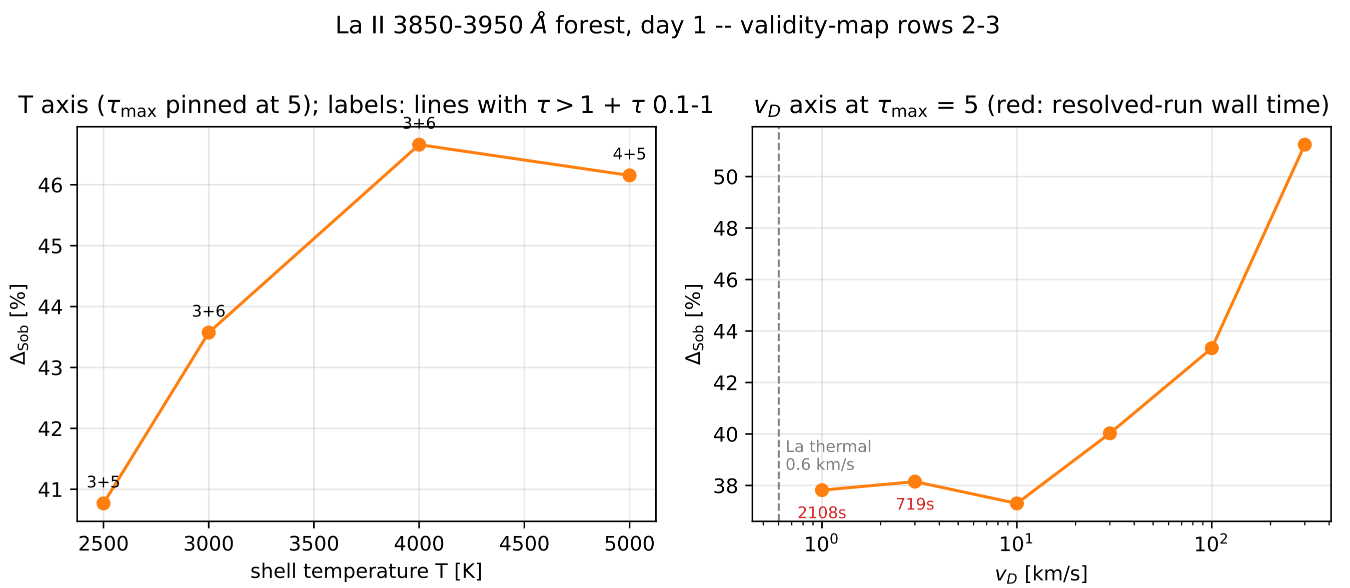

Figure 9. Left: temperature axis at pinned $\tau_{\max} = 5$ (weak dependence; annotations count strong lines). Right: the Doppler-width frontier: $\Delta_{\exp}$ is flat at $\sim 38\%$ from $10\,\mathrm{km\,s^{-1}}$ down to $1\,\mathrm{km\,s^{-1}}$, essentially the La thermal width; red annotations give the resolved-run wall time.

tion; those enter only through the optical depths they produce. The sweep holds the shell geometry and the Doppler width fixed, and §5.4 shows Δ does depend on v_D, so the collapse is onto $\tau_{\max}$ within a fixed (v_D, $\Delta v_{\rm shell}$) slice—not a claim that $\tau_{\max}$ alone determines the error everywhere. Nor is $\tau_{\max}$ the fundamental variable: §5.7 shows two forests of equal $\tau_{\max}$ can differ in $\Delta_{\exp}$ when one is blanketed and the other sparse. The quantity that controls $\Delta_{\exp}$ exactly is the transmission-weighted mean of e^D over rays, D the saturation deficit of Eq. (10) (Fig. 12); $\tau_{\max}$ tracks it because, at fixed geometry and width, the strongest few lines dominate D on the rays that still transmit. Both errors vanish together as $\tau_{\max} \to 0$—the weak-line limit both approximations share—and both grow with $\tau_{\max}$, but the closure departs by a median factor 8 more than the Sobolev localization wherever $\tau_{\max} > 1$.

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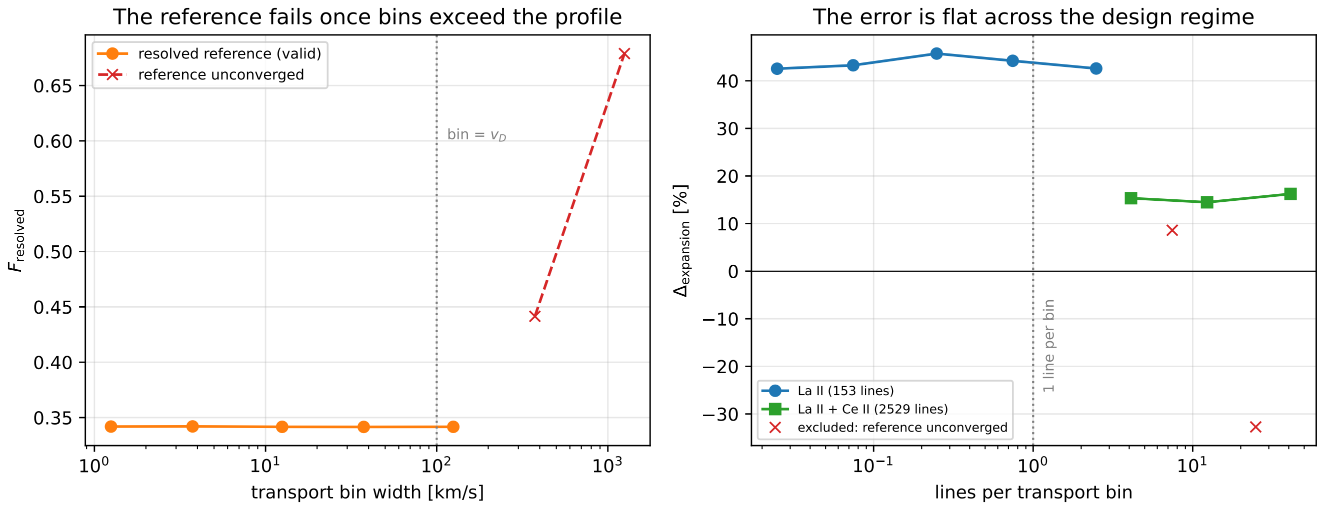

Figure 10. Left: the resolved reference is stable to 0.13% while the transport bin stays narrower than the line profile, and runs away once it does not—those points measure the reference failing, not the approximation changing, and are excluded. Right: across the valid range $\Delta_{\rm expansion}$ is flat from 0.025 to 41 lines per bin, two line lists and three decades.

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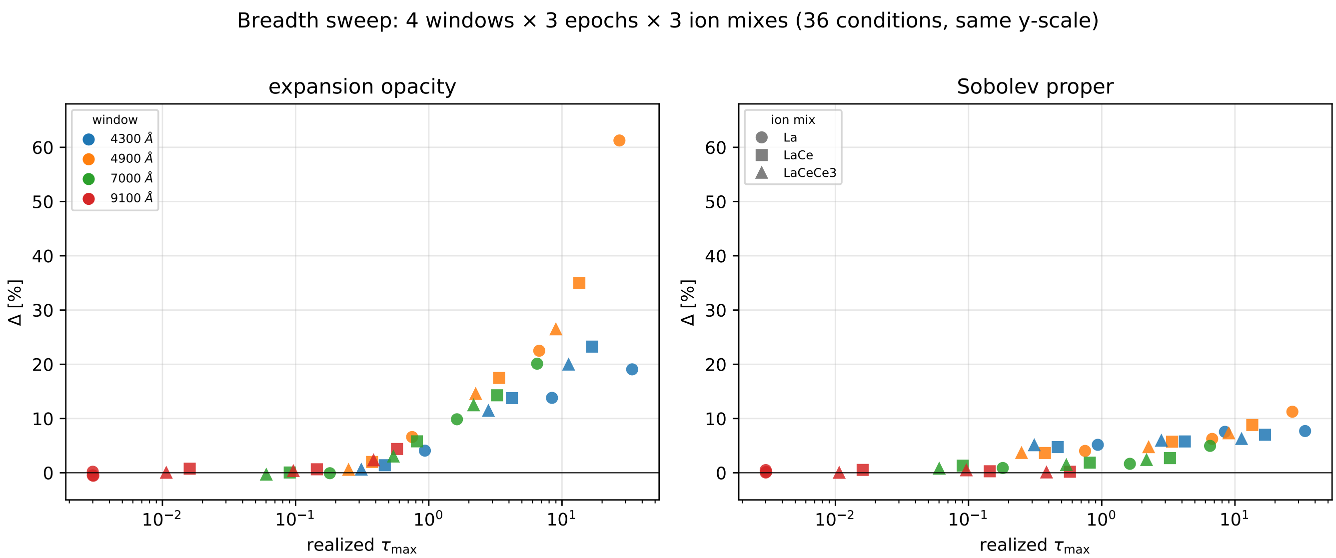

Figure 11. Thirty-six conditions—four wavelength windows (none of them the window of §5.3), three epochs, three ion mixtures—with both panels on the same vertical scale. $\Delta_{\rm exp}$ is a seed-matched SEDONA pair; $\Delta_{\rm Sob}$ is formed against the deterministic finite-profile reference on identical rays. Every condition follows one trend in the realized $\tau_{\rm max}$ within this fixed-geometry, fixed-width slice.

The 9100 Å window contains no strong La II lines ($\tau_{\rm max} \simeq 0$) and is kept as a null control: its band flux must be unity in every leg, and is. Every band flux in the sweep goes through the one normalization routine of §3.2, with the reference margin held clear of the final, partial bin of the transport grid.

5.7 Multi-ion blending and the density dependence

One axis remains: what happens when forests become dense enough that resonances stop being resolvable as individual features?

Adding Ce II at equal mass fraction multiplies the line count by 16 and drives strong-line separations down to $8\,{\rm km\,s^{-1}}$—genuine sub-Doppler blending. The measured error, however, *falls*, to $\Delta_{\rm exp} = +14.6\%$ (Fig. 13).

Finding F7: $\Delta_{\rm exp}$ is non-monotonic in forest density. With

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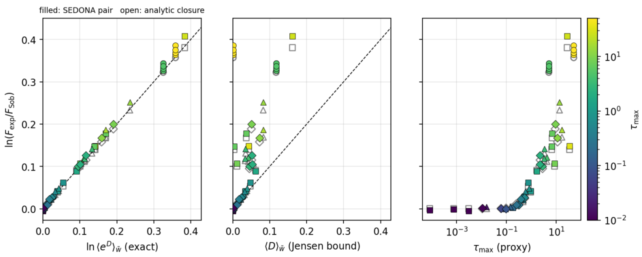

Figure 12. The controlling statistic. The closure's departure from the Sobolev leg, $\ln(F_{\rm exp}/F_{\rm Sob})$, for the 36 breadth conditions and the 12-point strength–width grid (filled: SEDONA pair; open: analytic closure) against (left) the exact predictor $\ln\langle e^D\rangle_{\tilde w}$, $\tilde w \propto e^{-S}$, with the 1:1 line—the analytic pair lies on it by identity, the SEDONA pair sits above it by the expansion-leg bin systematic of §5.5; (centre) the transmission-weighted mean deficit $\langle D\rangle_{\tilde w}$, a Jensen lower bound that fails once rays saturate; (right) $\tau_{\rm max}$, the empirical proxy.

enough crossings, even capped per-crossing contributions sum to blackness, so both treatments agree that almost nothing escapes and the relative error is suppressed.

F7 therefore defines the domain of relevance rather than weakening the result. The error is small in two limiting regimes—weak lines, and bands driven completely black—and largest in the *intermediate* one, where optically thick lines coexist with nonzero escaping flux. Partially transparent, feature-forming regions of exactly this kind do occur in realistic kilonova calculations. Multi-element models retain identifiable La, Ce and Sr features rather than becoming featureless (Domoto et al. 2022; Gillanders et al. 2022); three-dimensional line-by-line Sobolev calculations find spectra shaped by Sr II, Y II, Zr II, La III and Ce III together with continuum shaping by Ce and Nd (Shingles et al. 2023; Collins et al. 2023, 2026); and r-process opacity varies strongly with composition and by roughly an order of magnitude with epoch, so "lanthanide-rich" does not imply uniform blanketing at all wavelengths and times (Tanaka et al. 2020).

We are explicit about what this does and does not license. It does not license reading +38–48% as the error of a fully mixed kilonova spectrum; F7 shows directly that the error is suppressed as a band approaches complete blanketing, and our La II+Ce II case already demonstrates the suppression. What it does license is the statement that expansion opacity can generate tens-of-percent attenuation errors in strong, incompletely blanketed bands, and that such bands are not a contrived configuration.

5.8 Computational cost

The shortcut is used because of cost. On a standard SEDONA Type Ia example, switching from expansion opacity to the resolved treatment multiplied the run time by 378 (14 s against 88 min). Our sweep adds a cleaner measurement: in the controlled forest, resolved-run wall time scales as $1/v_D$ (33 s at 100 km s^{-1}, 719 s at 3 km s^{-1}, 2109 s at 1 km s^{-1}), driven entirely by the transport frequency grid—and the expansion run on the same fine grid costs the same (2305 s at 1 km s^{-1}). The expense is the grid, not the physics; narrow windows keep resolved calculations affordable, which is the strategy this program uses.

6 DISCUSSION

6.0.0.1 What the departure means for kilonova modeling. Lanthanide-rich ejecta are the definition of the strong-line regime: the opacity calculations that motivate this work find bound–bound optical depths in the line-forming region far above unity across the optical and near-infrared (Tanaka et al. 2020; Kato et al. 2024), and our sweep spans $\tau_{\rm max}$ up to 50, where the Poisson-versus-Bernoulli gap is already $\sim 45\%$ in band transmission and does not turn over at larger τ (Appendix A4). The expansion-opacity closure is exact in the statistic it was derived for—the expected number of line interactions per unit path, Karp's mean free path—and that is the statistic that governs diffusion and the random walk deep in the ejecta. What it does not reproduce is the deterministic attenuation of a resolved realization once individual lines saturate, and that is the statistic that governs how much continuum escapes through a strong-line band near the photosphere. Which of the two a calculation needs depends on the regime, and the difference between them is tens of percent wherever strong lines are sparse enough not to blanket.

6.0.0.2 One energy-conserving check. Everything above is the *attenuation of an externally supplied continuum*: radiative equilibrium is disabled (§3.2) and absorbed packet energy is not re-emitted. Nothing in a pure-absorption measurement bounds what happens once it is, because redistribution couples frequencies nonlinearly. We therefore ran the forest pair once more with SEDONA's radiative-equilibrium switch on, resolved against expansion, seed-matched, in two variants: a single iteration, which re-emits at the input temperature and isolates redistribution from population feedback, and an iterated run in which the gas temperature converges—to different values in the two modes, so that differential includes feedback. Three seed-matched pairs per variant, normalized on a blue margin because re-emission redshifted by up to 3000 km s^{-1} contaminates the red one (on the pure-absorption runs the two margins agree to 0.2–0.5%). In the emergent 3800–3955 Å band the resolved-versus-expansion differential is $+7.7 \pm 0.02\%$ after one iteration and $+5.0 \pm 0.7\%$ at convergence ($|\delta T/T| < 0.7\%$ over the last three iterations; median $T_{\rm gas}$ rises from 3000 to ~ 7700 K in both modes), against +44% in pure absorption on the same normalization. The band itself fills from 0.34 to 0.84–0.86 in both treatments, and the flux pure absorption removed reappears immediately redward, where the 3955–3995 Å band rises to 1.08–1.10 of the continuum; a $\tau_{\rm max} = 0.05$ control returns both modes to the continuum within 0.6%. So most of the closure's departure in attenuation does not survive into the emergent band once absorbed energy is re-emitted: the differential shrinks by a factor of six to nine, and its sign is preserved. We report this as one check on one configuration, not as the emergent-spectrum error budget, and we make no claim about the sign or size of the error in a fully mixed, multidimensional kilonova spectrum; the companion study addresses that.

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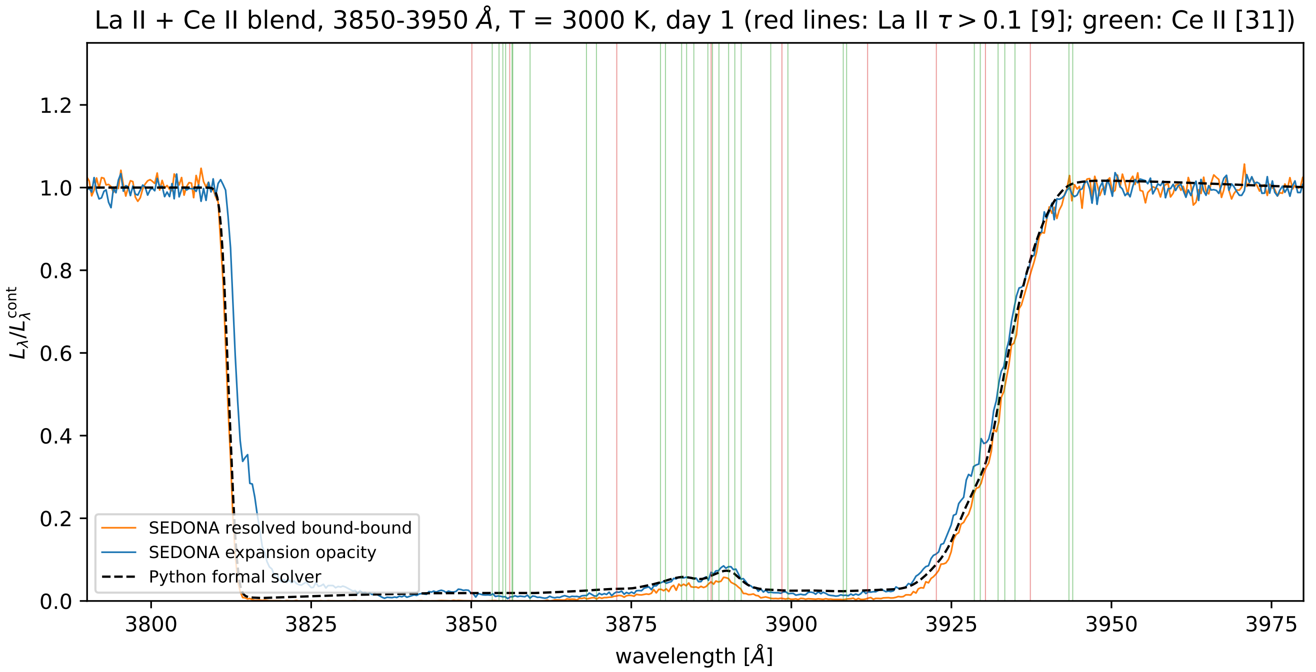

Figure 13. La II + Ce II blend in the same window. Ce floods it with 2376 additional lines (2529 total; strong-line spacings down to $8\,\mathrm{km\,s^{-1}}$), saturating the band to near-black in every treatment.

[t]

Table 3. The four results, and the subsidiary findings each collects.

#	Result	Where
R1	**The separation.** Against a deterministic finite-profile reference on identical rays, the Sobolev localization errs by $\lesssim 1\%$ at $v_D \le 30\,\mathrm{km\,s^{-1}}$ ($\tau_{\max} \le 5$) and $\lesssim 0.3\%$ at $\le 10\,\mathrm{km\,s^{-1}}$, while the expansion-opacity closure differs by +38–45% at every width; robust across 36 further windows, epochs and mixtures (formerly F6, F9, F10).	§5.3 §5.4 §5.6
R2	**The mechanism.** The closure preserves the expected interaction count $\sum(1-e^{-\tau_k})$ exactly and applies Poisson survival to a Bernoulli product; the departure is the saturation deficit $\sum[\tau_k-(1-e^{-\tau_k})]$ per ray, so it is per-resonance, width-independent, bin-width invariant, and maximal for strong sparse forests (formerly F4, F5, F7, F13).	§2.5 App. A4 §5.5 §5.7
R3	**The Sobolev error is geometric.** Overlap is inert in pure absorption (optical depths add); the residual is profile clipping at the edges of the finite line-forming region, $\propto v_D/\Delta v_{\rm shell}$, and symmetric gradients cancel the locality error at leading order (formerly F1, F12).	App. A6 §4
R4	**Three conventions must be matched** in any cross-code comparison: thermal emission, spectral normalization, and transport treatment (frozen snapshot $\tau/\tau_{\rm S}=(1-\beta)/\gamma$ vs worldline $1/\gamma$); each produces few-percent offsets of the size under study, which is why the Sobolev differential is formed against a deterministic reference (formerly F3, F8, F11).	§3.6 App. A7 §3

6.0.0.3 What exactly was measured on the Sobolev side. Our Sobolev leg is the approximation in its pure form—$e^{-\tau_{\rm S}}$ per line, resonances treated as planes—evaluated analytically. That evaluation is *exact* rather than approximate, but only because the configuration is pure absorption without scattering: it is precisely that restriction which makes the p-averaged staircase equal to per-line Sobolev transport. A per-line Sobolev scheme inside a scattering Monte Carlo code—the macroatom treatment of TARDIS-class codes (Lucy 2002, 2003; Kerzendorf & Sim 2014), say—is a different object, and our numbers should not be read as bounding it; extending the harness there is the natural sequel, and is the regime in which the leg would have to be built rather than computed.

Within that scope the attribution is clean. The per-crossing cap (F4, F5) is absent from the Sobolev leg by construction and present in expansion opacity by construction, and the measured errors follow suit. Of the two Sobolev-intrinsic assumptions, neither is stressed here: locality is not, because our populations are constant (and F1 shows it would cancel at leading order even if it were), and isolation cannot be, because in pure absorption optical depths add exactly regardless of overlap. What remains is the geometric boundary effect of Appendix A6, which is a property of the finite line-forming region rather than of the approximation.

6.0.0.4 Limitations. Pure absorption with LTE populations (no scattering, fluorescence, or NLTE); fixed temperatures; a controlled homologous test flow at 1000–3000 $\mathrm{km\,s^{-1}}$ using kilonova line lists and thermodynamic conditions, not the 0.1–0.3c of real ejecta (Appendix A7 sets out what changes there, and the machinery now reaches it, but no measurement in this paper is made at those velocities); Doppler widths down to 1 $\mathrm{km\,s^{-1}}$ in the deterministic reference and 1 $\mathrm{km\,s^{-1}}$ in SEDONA, against the 0.6 $\mathrm{km\,s^{-1}}$ thermal value; two elements and three ionization stages across four windows and three epochs, but no full lanthanide mixture; Monte Carlo noise of 0.26–0.30% per single-realization band flux, reduced by seed averaging where quoted (§3.5); and, apart from the single check above, radiative equilibrium disabled, so the reported fluxes are the attenuation of an externally supplied continuum rather than an emergent spectrum. The Sobolev localization is tested with its two assumptions deliberately protected—locality by constant populations, isolation by pure absorption, in which optical depths add regardless of overlap—so the statement licensed is the narrow one: for fixed populations, pure absorption, homologous expansion and narrow intrinsic profiles, replacing each finite profile by a localized resonance introduces only the boundary error of Appendix A6. Whether that holds once populations respond to the radiation field, and whether the closure's departure survives radiative redistribution, are the subjects of the companion study.

7 CONCLUSIONS AND NEXT STEPS

We built and validated a harness that isolates the line-transfer treatment in expanding ejecta—a deterministic finite-profile reference, a per-line Sobolev leg and the expansion-opacity closure evaluated on identical rays, with a Monte Carlo code as the independent validation—and used it to measure, on calibrated lanthanide data, two approximations that the literature carries under a single name.

They preserve different properties of a line forest. The expansion-opacity closure of Karp et al. (1977) and Eastman & Pinto (1993) is exact in the statistic it was built for: the expected number of line interactions per unit path, $\sum_k(1-e^{-\tau_k})$, is reproduced identically at every bin width. What it does not reproduce is the deterministic transmission of a resolved line realization, because it applies the survival of a Poisson process to what is a product of Bernoulli survivals; the two agree while every line is weak and separate, per resonance and by the saturation deficit $\sum_k[\tau_k-(1-e^{-\tau_k})]$, once lines saturate. On the La II forest that separation is +38% in band transmission at thermal line widths and at every bin width from 0.025 to 41 lines per bin, rising to +45% in the Monte Carlo implementation at 100 $\mathrm{km\,s^{-1}}$; across 36 further conditions it is robust, with the transmission-weighted saturation deficit as the controlling statistic and the strongest line's optical depth as its proxy within a fixed geometry. It is largest not in fully blanketed bands, where both treatments agree that little escapes, but in the intermediate regime where strong lines have not yet merged—the regime in which realistic calculations still show identifiable spectral features (Domoto et al. 2022; Tanaka et al. 2020). The Sobolev localization itself, measured against the deterministic reference, carries no strength floor: its error is the clipping of profiles at the edges of the finite line-forming region, $\lesssim 1\%$ at $v_D \le 30\,\mathrm{km\,s^{-1}}$ for $\tau_{\max} \le 5$ and $\lesssim 0.3\%$ at $v_D \le 10\,\mathrm{km\,s^{-1}}$, vanishing at thermal widths (Appendix A6).

The practical implication for kilonova modelling is that "we use the Sobolev approximation" and "we use expansion opacity" are not interchangeable statements: the second is exact for the mean free path and carries a quantified, strength-set departure from resolved transmission wherever strong lines are sparse. Which statistic a calculation needs is a question about its regime, and it deserves to be asked explicitly.

Next: worldline-consistent comparison at realistic ejecta velocities (0.1–0.3c), which SEDONA's frozen steady-state mode cannot supply—though its time-dependent mode can, and is the natural route; and extension to scattering and fluorescence, which is where the analytic equivalence exploited here breaks down and a per-line Sobolev *transport* scheme must be built and tested rather than computed in closed form.

REPRODUCIBILITY

All code, data provenance, executed notebooks, experiment generators, and this manuscript live at `https://github.com/zhuygln/revisit_sobolev`. Atomic data: GSI Database for Kilonova Radiative Transfer, Zenodo record 19335084 (CC-BY 4.0). Radiative transfer: the public SEDONA release (`https://github.com/dnkasen/pubsed`). Every figure regenerates from a committed script; the 68-test suite pins the physics of every module.

APPENDIX A: DERIVATIONS FOR THE NON-SPECIALIST

A1 The formal solution and thermal populations

Multiply Eq. (1) by e^{τ} with $\mathrm{d}\tau = \alpha\,\mathrm{d}s$ and integrate: the intensity leaving a slab is

$$I^{\mathrm{out}} = I^{\mathrm{in}} e^{-\tau_{\mathrm{tot}}} + \int_0^{\tau_{\mathrm{tot}}} S(\tau')\, e^{-(\tau_{\mathrm{tot}}-\tau')}\,\mathrm{d}\tau', \qquad S \equiv j/\alpha. \tag{A1}$$

The first term is the attenuated input beam; the second adds the gas's own emission, each parcel attenuated by the material still in front of it. The deterministic solver evaluates Eq. (A1) numerically along each ray with $S = B_\nu(T)$, accumulating τ by the trapezoidal rule; convergence requires resolving the Doppler width along the path.

In thermal equilibrium the fraction of ions in level i (energy E_i, statistical weight $g_i = 2J_i + 1$) is

$$\frac{n_i}{n_{\mathrm{ion}}} = \frac{g_i\, e^{-E_i/k_B T}}{\sum_j g_j\, e^{-E_j/k_B T}}. \tag{A2}$$

A subtlety our tests caught: the *fraction per statistical weight* is what decreases monotonically with energy; raw fractions need not (at 3000–5000 K a $g = 7$ level at 1016 $\mathrm{cm^{-1}}$ outpopulates the $g = 5$ ground state of La II).

A2 The Sobolev optical depth

Along a ray toward the observer, the gas at coordinate z moves with line-of-sight velocity z/t (homologous flow), so a photon of observer frequency ν has gas-frame frequency $\nu' = \nu\,(1 - z/ct)$ to first order in v/c. Insert the line opacity (Eq. 5) into the optical-depth integral (Eq. 2) and change variables from z to ν' ($\mathrm{d}\nu' = -(\nu/ct)\,\mathrm{d}z \approx -(\nu_0/ct)\,\mathrm{d}z$ near resonance):

$$\tau = \sigma_{\rm cl} f \int n_l(z)\,\phi(\nu' - \nu_0)\,\mathrm{d}z \approx \sigma_{\rm cl} f\, n_l(z_{\rm res})\,\frac{ct}{\nu_0}\underbrace{\int \phi\,\mathrm{d}\nu'}_{=1} = \sigma_{\rm cl} f\, n_l\,\lambda_0\,t, \quad (\mathrm{A3})$$

which is Eq. (7). The two approximations made are exactly the two Sobolev assumptions: n_l was pulled out of the integral at its resonance-point value (locality), and the profile was integrated as if no other line interfered (isolation).

A3 Why a symmetric gradient cancels (F1)

Let $n_l(v) = n_0[1+g(v-v_{\rm res})]$ with g odd about the resonance (e.g. tanh centered there). The resolved integral weights n_l by the even profile ϕ; the odd term integrates to zero, so the first-order error vanishes. The next term is set by the curvature $n_l''(v_{\rm res})$—which for a centered tanh *also* vanishes (inflection point). Only with the resonance off-center does the curvature term survive, giving $E_{\rm Sob} \simeq \frac{1}{2}|n_l''/n_l|\,\sigma_v^2 \propto \epsilon^2$, the law measured in Fig. 1.

A4 What expansion opacity preserves, and what it does not (R2)

Integrate the binned opacity of Eq. (8) along the stretch of path over which the moving gas keeps one line inside one frequency bin. The Doppler sweep rate ties path length to bin width, $\Delta z = ct\,\Delta\nu_{\rm bin}/\nu$, so the contribution of line k is

$$\alpha^{\rm exp}\,\Delta z = \frac{1}{ct}\,\frac{\nu_k}{\Delta\nu_{\rm bin}}\left(1 - e^{-\tau_{{\rm S},k}}\right) \times \frac{ct\,\Delta\nu_{\rm bin}}{\nu_k} = 1 - e^{-\tau_{{\rm S},k}}. \quad (\mathrm{A4})$$

The bin width cancels. Summed over the resonances a ray crosses, $\int \alpha^{\rm exp}\,\mathrm{d}s = \sum_k (1 - e^{-\tau_k}) \equiv E$: the expected number of line interactions for a photon that is counted but not removed, which is the mean-free-path statistic Karp et al. (1977) built the coefficient from. The closure reproduces E exactly, at any resolution.

Transmission is e^{-S}, $S = \sum_k \tau_k$ (a product of Bernoulli survivals); a code that exponentiates $\int \alpha^{\rm exp}\mathrm{d}s$ returns e^{-E} (Poisson survival with the same mean). They agree when every $\tau_k \ll 1$ and separate by the saturation deficit $D = S - E \geq 0$ per ray (Eq. 10)—a property of E and S, not of the grid, which is why the departure is per-resonance, width-independent and bin-width invariant.

For a band average over core rays with weights w (the $p\,\mathrm{d}p$ measure of Appendix A5) the statement is an identity,

$$\frac{F_{\rm exp}}{F_{\rm Sob}} = \frac{\langle e^{-E}\rangle_w}{\langle e^{-S}\rangle_w} = \frac{\langle e^{-S}\,e^{D}\rangle_w}{\langle e^{-S}\rangle_w} = \langle e^{D}\rangle_{\tilde{w}}, \qquad \tilde{w} \propto w\,e^{-S}, \quad (\mathrm{A5})$$

the transmission-weighted mean of e^D, pinned in the test suite to 10^{-13}. Its content is physical rather than algebraic: the rays that still transmit are weighted most, and on those the strongest crossed lines dominate D. That is why the realized $\tau_{\max}$ is a good proxy within a fixed geometry (§5.6), and why the naive unweighted deficit $\langle D\rangle$ is not—by Jensen's inequality $\ln\langle e^D\rangle_{\tilde{w}} \geq \langle D\rangle_{\tilde{w}}$, and once rays saturate the two separate widely (Fig. 12).

A5 Averaging over the source disk (F5's technical companion)

A resonance plane at height $z_{\rm res}$ in front of a core of radius $r_{\rm core}$ is crossed by the core ray at impact parameter p only if $\sqrt{r_{\rm core}^2 - p^2} < z_{\rm res} < \sqrt{r_{\rm out}^2 - p^2}$. The observed attenuation is therefore

$$\frac{F}{F_{\rm cont}}(\nu) = \frac{2}{r_{\rm core}^2}\int_0^{r_{\rm core}} p\,\exp\Big[-\sum_{k\ {\rm crossed}(p,\nu)} \tau_k\Big]\,\mathrm{d}p, \quad (\mathrm{A6})$$

not $e^{-\sum\tau}$ with a global crossing criterion: planes at $z_{\rm res} < r_{\rm core}$ shadow only part of the disk. Ignoring this (∼plane counting) predicts measurably too-shallow troughs in the ladder experiment.

A6 Boundaries, not overlap: where the Sobolev error comes from

Two Sobolev-intrinsic assumptions could in principle fail: locality and isolation. Neither produces the residual measured in §5.3.

A6.0.1 Isolation cannot fail here. With fixed populations and pure absorption the opacities of different lines simply add, so

$$\tau_\nu = \int \sum_i \alpha_{\nu,i}\,\mathrm{d}s = \sum_i \int \alpha_{\nu,i}\,\mathrm{d}s = \sum_i \tau_{\nu,i}, \quad (\mathrm{A7})$$

and the emergent intensity is $I_{\nu,0}\exp(-\sum_i \tau_{S,i})$ whether the profiles are separated or lie exactly on top of one another. Driving two identical lines from 20 Doppler widths apart to exact coincidence reproduces the Sobolev result to six decimal places at every separation. Overlap becomes a real effect only when scattering, fluorescence or NLTE populations allow one transition to change the radiation field another one samples; those are switched off here by construction.

A6.0.2 What does fail is geometric. The line-forming region is finite. A resonance at $z_{\rm res}$ in a shell spanning $z_{\rm lo}$ to $z_{\rm hi}$ has its profile clipped by the edges, so for a central ray

$$\frac{\tau_{\rm resolved}}{\tau_{\rm S}} = \frac{1}{2}\left[\mathrm{erf}\left(\frac{z_{\rm hi} - z_{\rm res}}{\ell_{\rm S}}\right) - \mathrm{erf}\left(\frac{z_{\rm lo} - z_{\rm res}}{\ell_{\rm S}}\right)\right], \quad (\mathrm{A8})$$

with the Sobolev length $\ell_{\rm S} = v_D t$. This is unity deep inside, exactly 1/2 at an edge and zero outside—while the Sobolev prescription applies a step of 1 inside and 0 outside. Direct integration reproduces this to four decimals at every width tested (Fig. A1, left), and the curves collapse when plotted against d/v_D, confirming that the width enters only through the ratio.

The band-averaged consequence follows: the affected fraction of the spectrum is the fraction of the velocity span lying

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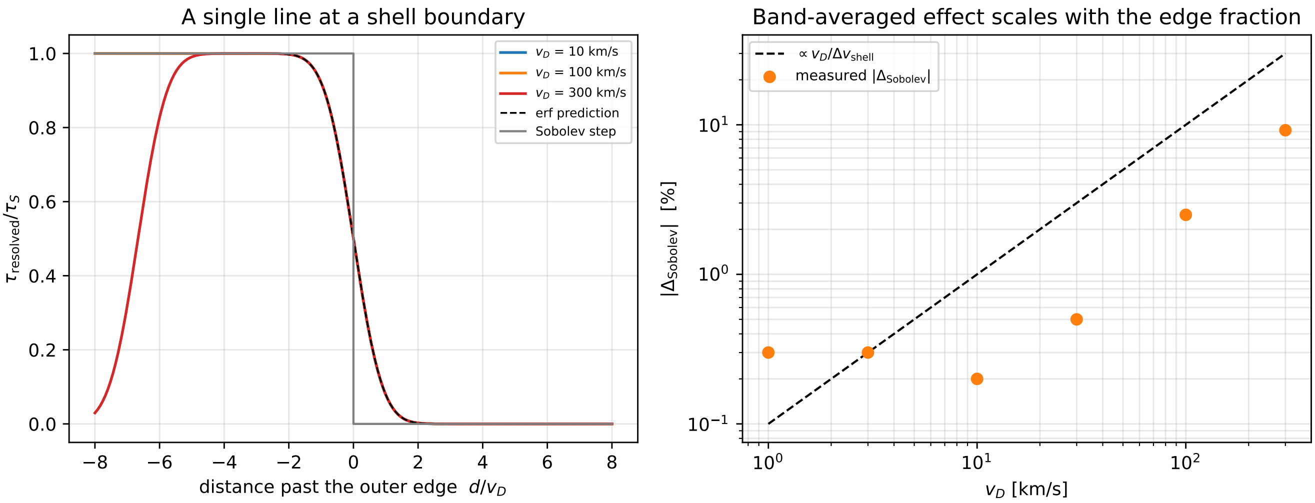

Figure A1. Left: a single line's resolved optical depth as its resonance is walked out through the outer edge of the line-forming region, in units of the Doppler width. The curves for $v_D = 10$, 100 and $300\,\mathrm{km\,s^{-1}}$ coincide — the effect is self-similar in d/v_D — and follow the error-function prediction exactly, against the hard step that Sobolev applies. The $300\,\mathrm{km\,s^{-1}}$ curve departs from unity on the left because the shell is then only ~ 7 Doppler widths wide and the inner edge clips as well. Right: the band-averaged consequence tracks $v_D/\Delta v_{\rm shell}$.

within a few Doppler widths of an edge. With two edges that is $\sim 2v_D/\Delta v_{\rm shell}$, so for our shell ($\Delta v_{\rm shell} = 2000\,\mathrm{km\,s^{-1}}$) it runs from 1% at $v_D = 10\,\mathrm{km\,s^{-1}}$ to 30% at $300\,\mathrm{km\,s^{-1}}$. This is an upper bound on the band error rather than a prediction of it, since only the clipped part of each profile is misassigned: the measured $|\Delta_{\rm Sob}|$ over the same range is 0.2% and 9.2% (Fig. A1, right). What the estimate does capture, and what matters here, is the linear scaling in $v_D/\Delta v_{\rm shell}$ and hence the two orders of magnitude by which the effect shrinks at physical line widths.

This is worth stating plainly because it is easy to mistake for physics: the error is a property of imposing a *finite* line-forming region on a calculation whose approximation assumes a locally infinite one. In a real ejecta the same effect appears at the photosphere and at the outer edge of the line-forming zone, and it scales with the thermal width—which for lanthanides is $\sim 0.6\,\mathrm{km\,s^{-1}}$ against ejecta spans of $10^4\,\mathrm{km\,s^{-1}}$, i.e. utterly negligible. Our sensitivity to it at $v_D = 100$–$300\,\mathrm{km\,s^{-1}}$ is an artifact of using artificially broad lines to keep the frequency grids affordable.

A7 Frozen snapshot versus photon worldline: what F3 really was

Appendix A2 derived τ_S to first order in v/c. Kilonova ejecta move at 0.1–0.3c, and the solver measures that a resonance at $z_{\rm res}/ct = 7.7\%$ gives a trough matching $\exp[-\tau_S(1 - z_{\rm res}/ct)]$ to four digits rather than $\exp(-\tau_S)$. The natural reading is that this is the leading relativistic correction and that it is $\mathcal{O}(v/c)$. It is not; it is the law of a frozen snapshot, and the distinction between the two is a transport treatment that must be matched across codes.

A7.0.1 Two treatments. The $\mathcal{O}(v/c)$ law follows from differentiating the Doppler factor $D = \gamma(1 - b)$ *at fixed* t, i.e. from pushing a photon through a frozen snapshot of the ejecta. But a photon takes time to cross a resonance, and in homologous flow $\boldsymbol{\beta} = \mathbf{r}/(ct)$ depends on the age as well as the position. Along the photon worldline $\mathbf{r}(t) = \mathbf{r}_0 + c\hat{\mathbf{n}}(t - t_0)$,

$$\frac{\mathrm{d}\boldsymbol{\beta}}{\mathrm{d}t} = \frac{\hat{\mathbf{n}} - \boldsymbol{\beta}}{t} \qquad \Longrightarrow \qquad \frac{\mathrm{d}b}{\mathrm{d}t} = \frac{1 - b}{t}, \tag{A9}$$

whereas the frozen calculation uses $\mathrm{d}b/\mathrm{d}z = 1/(ct)$. The two differ by a factor $(1 - b)$—first order in β, the same order as the effect being computed. The time term does not vanish because the crossing is brief: what matters is the *gradient*, and the time term is proportional to b itself.

A7.0.2 The correct law. With $D = \gamma(1 - b)$ one finds $\mathrm{d}D/\mathrm{d}t = -D\gamma^2(1 - b)/t$. Using the opacity transformation $\chi_{\rm lab} = D\chi'$ (from the invariance of $\nu\chi_\nu$) and the resonance condition $D\nu = \nu_0$, the profile integral collapses to $\tau = D\chi' c/|\mathrm{d}\nu'/\mathrm{d}t|$, and the factors of $\gamma(1 - b)$ cancel down to a single Lorentz factor:

$$\boxed{\frac{\tau_{\rm rel}}{\tau_S} = \frac{1}{\gamma} = \sqrt{1 - \beta^2}} \tag{A10}$$

with τ_S evaluated using the density and ejecta age local to the resonance, which makes the statement independent of how the calculation is anchored in time. **There is no first-order term**: $1/\gamma = 1 - \frac{1}{2}\beta^2 + \mathcal{O}(\beta^4)$. The frozen calculation instead gives, for radial rays, $\tau_{\rm frozen}/\tau_S = (1 - \beta)/\gamma$.

A7.0.3 Numerical confirmation. Integrating the resolved line opacity directly along the path—no Sobolev assumption anywhere—reproduces both expressions to five decimal places, with the ejecta age held fixed or allowed to advance:

So the relativistic correction at $\beta = 0.1$, 0.2, 0.3 is 0.5%, 2.0%, 4.6%—not the 10%, 22%, 33% that the frozen law implies.

β	frozen $\tau/\tau_{\rm S}$	worldline $\tau/\tau_{\rm S}$	$1/\gamma$
0.05	0.94881	0.99875	0.99875
0.10	0.89549	0.99499	0.99499
0.20	0.78384	0.97980	0.97980
0.30	0.66776	0.95394	0.95394

A7.0.4 Which problem does the Monte Carlo code solve? It is natural to assume a Monte Carlo transport code supplies the time-dependent answer. It does not, in the configuration used here. Sweeping a single line's resonance velocity from $\beta = 0.001$ to 0.34 and comparing the measured $\tau_{\rm eff} = -\ln(F/F_{\rm cont})$ against both laws gives an RMS deviation over $\beta_{\rm res} \gtrsim 0.1$ of 0.024 from the frozen law and 0.55 from the worldline law—a factor 23—and at $\beta_{\rm res} = 0.34$ the measurement is 0.613 against a frozen prediction of 0.618 and a worldline prediction of 1.43. SEDONA run with `transport_steady_iterate` iterates the radiation field on a fixed grid at fixed epoch; its source disables hydrodynamics in that mode and reads no time-stepping parameters. It is a frozen-snapshot calculation.

The code itself is not restricted to that. Left in its time-dependent mode it advances the grid homologously—diluting the density as t^{-3} and expanding the boundary—while photons still in flight are carried across timesteps, which is the worldline problem discretized at timestep resolution. We did not exercise that mode for the measurements reported here, and the reader should read the comparison above as a property of the configuration rather than of the code.

This matters for the harness. Cross-code comparison requires matching the *transport treatment* as well as the thermal-emission convention (F8) and the spectral normalization (§5.6): a third convention, and the one with the largest lever arm at high velocity. Like-for-like against a frozen SEDONA means running the solver in its frozen mode. At the shell velocities used for the measurements in this paper ($\beta \leq 0.01$) the two laws differ by $\lesssim 2\%$ in τ. That is far below the level at which any $\Delta_{\rm exp}$ moves—those are same-code differentials—but it is comparable to the $\sim 1\%$ resolved-leg agreement itself, so every solver-versus-SEDONA comparison in this paper is quoted with the solver in frozen mode, matching SEDONAs transport treatment. At kilonova velocities the choice would dominate.

A7.0.5 What the $\mathcal{O}(v/c)$ offset is. Not a frame ambiguity and not the leading relativistic correction, but the law of a frozen snapshot—a problem the photon does not solve, and the one a steady-iterate transport calculation does. The distinction earns its own entry in Table 3 as F11, because "neglect of light-travel-time evolution" is a separate approximation from the Sobolev localization, from the independent-line assumption, and from expansion-opacity binning. Its practical consequence for the measurements reported here is that at our shell velocities the correct correction is 5×10^{-5}: negligible, and therefore *not* an explanation for the Sobolev residual of §5.3.

For the general case the resonance locus is not a plane—γ depends on r, not z—and one needs the common-direction and common-point frequency surfaces of Jeffery (1993). Our controlled experiments keep the emitting core small compared with the resonance radii, so all core rays are nearly radial and the plane picture holds.

A7.0.6 What changes at realistic velocities. The machinery now reaches 0.1–0.3c, and two things learned in building it bear on any future measurement there, though none is made in this paper. First, the worldline correction $1/\gamma$ is not the controlling term. A photon leaving the source at t_0 meets the material moving at β at $t_{\rm res} = t_0/(1-\beta)$—later than the frozen-geometry estimate $t_0(1+\beta)$, because the fluid element is itself moving outward while the photon chases it—and with $n \propto t^{-3}$ and $\tau_{\rm S} \propto nt$ the medium contributes $(1-\beta)^2$ against transport's $1/\gamma$: $\tau/\tau_{\rm S}(t_0) = (1-\beta)^2/\gamma$, confirmed against direct integration of the resolved opacity with the density evolving along the path. At $\beta = 0.3$ the $1/\gamma$ deficit is 4.6% and the light-travel dilution is 51%; a calculation that advances the kinematics but not the medium is wrong by an order of magnitude more than the term it keeps. Second, under worldline transport the resonance locus is linear in z—with $ct(z) = z + Z_0$, $D = Z_0/\sqrt{Z_0^2 + 2Z_0 z - p^2}$, so $D = \nu_0/\nu$ has one root, $z_{\rm res} = \frac{1}{2}Z_0[(\nu/\nu_0)^2 - 1] + p^2/2Z_0$—and $\tau/\tau_{\rm S}(t_{\rm res}) = 1/\gamma$ exactly for every impact parameter. The two-root quadratic behind the common-direction and common-point surfaces of Jeffery (1993) belongs to the frozen snapshot; in the physical treatment the geometry objection does not arise. (For the special-relativistic form of the expansion opacity itself see Li 2019.) A pilot on a synthetic forest with every controlling ratio held fixed finds the Sobolev localization error independent of β and the closure's departure growing with it; it is a pilot, on a synthetic forest, and is not quoted as a result.

APPENDIX B: RELATION TO PREVIOUS LINE-OPACITY TREATMENTS

B1 Expansion opacity and line-binned alternatives

The problem addressed here sits inside a longer effort to represent extremely dense line forests without resolving every intrinsic profile. The Karp et al. (1977) and Eastman & Pinto (1993) construction converts each transition's Sobolev interaction probability $1 - e^{-\tau_{\rm S}}$ into an effective opacity over a frequency interval, replacing an enormous set of narrow lines by a tractable coarse-frequency coefficient. It is on that convenience that its wide adoption rests.

Fontes et al. (2020) approached the same computational problem differently. Their line-binned opacity preserves the frequency-integrated area of the underlying fine-structure opacity and can be tabulated independently of the velocity field. They wrote the expansion-opacity optical depth per bin as $\sum_i (1 - e^{-\tau_i})$ against the line-binned $\sum_i \tau_i$, noted that the former limits the contribution of a single line to one while using the full opacity at low optical depth, and compared both against a continuous Monte Carlo Sobolev treatment—which they describe as the more accurate one—finding broadly similar light curves and spectra, with the agreement explained by the similarity of the opacities in the low-optical-depth limit where transport matters in their ejecta. The unit cap of a single line in the expansion-opacity form is therefore theirs to state, and we do not claim it.

The present work is not another prescription; its purpose is diagnostic. What it adds is a finite-profile resolved calculation inserted *ahead of* per-line Sobolev transfer, so that the hierarchy resolved → Sobolev proper → expansion opacity lets an error in the final calculation be assigned either to

Sobolev localization or to the closure built on top of it; a controlled map of each step's error against intrinsic width, line strength, bin width, line density and calibrated lanthanide line lists; and the identification of the ensemble statistic—the transmission-weighted saturation deficit—that controls the closure's departure. That assignment and that map, rather than any individual band flux, are the contribution.

B2 Line overlap as a distinct failure mechanism

A separate limitation concerns whether successive line interactions remain independent. Banerjee et al. (2022) introduced a critical opacity corresponding to the condition that the characteristic line spacing becomes comparable to the intrinsic thermal width, and found far-ultraviolet expansion opacities of highly ionized lanthanides well above it at early epochs, implying substantial profile overlap. Banerjee et al. (2024) extended the heavy-element calculations and likewise emphasized that expansion opacity can underestimate the true opacity once neighbouring resonance regions overlap, because free propagation between independent interactions is then no longer a good description.

That mechanism is physically distinct from the one isolated here. Our production calculations use fixed populations and pure absorption, so $\alpha_\nu = \sum_i \alpha_{\nu,i}$ and hence $\tau_\nu = \sum_i \tau_{\nu,i}$ exactly, even when profiles coincide. The controlled coincidence experiment of §5.3 confirms this numerically to six decimal places. Our measured expansion-opacity floor therefore does *not* require any breakdown of line independence—which is worth stating, because it would be easy to attribute the two effects to one cause. Overlap becomes dynamically consequential once scattering, fluorescence or NLTE populations let one transition modify the radiation field another samples, and that lies outside the present scope.

B3 Line-by-line Sobolev transport and fluorescence

A complementary direction avoids binning altogether and propagates packets through individual Sobolev resonances. Modern calculations now perform line-by-line Sobolev transfer for tens of millions of bound–bound transitions in multidimensional merger ejecta, following fluorescence and associating individual spectral structures with particular ions (Shingles et al. 2023; Collins et al. 2023, 2026).

Those calculations answer a different question: what spectra emerge from increasingly realistic ejecta once line-by-line Sobolev transport is adopted. We ask instead whether the Sobolev replacement is itself accurate relative to a finite-profile calculation, and how much further error binning introduces. The two are complementary in a specific and useful way: our finding that the per-line Sobolev leg tracks resolved transfer to $\lesssim 0.5\%$ at thermal-like widths is a controlled justification for adopting line-by-line Sobolev transport as the reference treatment in exactly such calculations.

B4 Recent scrutiny of the Eastman–Pinto prescription

Morag (2026) revisits the physical interpretation of the Eastman & Pinto (1993) prescription in supernova transfer, showing that it can capture the photon mean free path through a line forest while substantially underestimating photon emissivity and reprocessing rates when used as a coarse-frequency emissivity. The broader point is that an effective opacity need not reproduce every property of the ensemble it replaces.

Our result is the same point from the absorption side, made exact. The coefficient preserves the expected interaction count per unit path—the mean free path Morag finds it captures—identically; what it does not preserve is the transmission of a deterministic line realization, because Poisson survival and a Bernoulli product differ once lines saturate (Appendix A4). We do not test an emissivity closure; apart from the single radiative-equilibrium check of §6, we measure the attenuation obtained when the expansion-opacity coefficient is inserted into otherwise identical transport, against resolved and per-line Sobolev calculations of the same configuration. The bin-width experiment of §5.5 matters in this context, because it shows the discrepancy persisting from well under one line per bin to tens of lines per bin while the resolved reference stays converged. The result therefore does not depend on using a statistical opacity in a numerically isolated-line regime. How much of this attenuation-level difference survives once absorbed packets are reprocessed and energy is conserved is left, deliberately, to the companion calculation described in §6.

B5 Atomic-data uncertainty versus transport uncertainty

Improvements in r-process atomic data define another, largely independent, uncertainty budget. Kato et al. (2024) find that improved singly ionized lanthanide calculations alter individual-element opacities by factors of several and mixture opacities appreciably; the calibrated GSI calculations used here supply experimentally anchored wavelengths for a large fraction of the strong transitions (Flörs et al. 2026b); and Fontes et al. (2026) show that changing the neodymium dataset alone moves kilonova spectra substantially and the peak bolometric luminosity by nearly a factor of 1.5.

These results motivate rather than confound the present experiment. We hold the atomic dataset fixed across every transport leg, so the differences reported here are not an estimate of the total uncertainty in kilonova modelling. They isolate a transport-formalism uncertainty that exists *in addition* to the atomic-physics uncertainty now being actively quantified—and which, unlike that uncertainty, will not be removed by better line lists.

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